Climate and Conservation Finance Feasibility Study

Natural Climate Solutions |

October 2019

Mississippi Headwaters

THE TRUST FOR PUBLIC LAND

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Cover photo: Crow Wing State Forest
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Senate
State of Minnesota

June 13, 2018

Peggy Ladner
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The Nature Conservancy
Minnesota, South Dakota and North Dakota
1101 West River Parkway, Suite 200
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Will Abberger
Director, Conservation Finance
The Trust for Public Land
306 N. Monroe Street
Tallahassee, FL 32301

Dear Ms. Ladner –

The Senate Environment and Natural Resources Policy and Legacy Finance Committee requests technical advice and assistance from The Nature Conservancy in connection with our efforts to develop and sustain reliable, ongoing sources of funding for healthy water protection in Minnesota.

As part of your advice and assistance, I understand you will conduct research with affected groups to develop recommendations for the Committee on strategies for supporting sources of public funds for protection and restoration of our healthy waters.

We are interested not only in the factual information that you can provide, but also your opinions and recommendations on public funding measures available to us and strategies to enact such measures. That would include understanding the public's priorities and attitudes concerning land preservation, water quality and natural areas, and how the public would respond to different methods of presenting those issues.

Information provided to the Committee will be public record as a matter of law. This request does not in any way commit public funds to the efforts of The Nature Conservancy and The Trust for Public Land related to this request, nor does it require public disclosure of any confidential information of either organization.

This request will continue in effect for any advice you offer or presentations you submit for the use of this body related to such matters. In addition, we would like to take this opportunity to request that you continue to be available to provide technical advice and assistance in this area and on related matters in the future.

Sincerely,

Senator Carrie Ruud
Minnesota State Senate – District 10

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INTRODUCTION

The Trust for Public Land (TPL) has been connecting communities to the outdoors, and to each other, since 1972. Our mission is to create parks and protect land for people, ensuring healthy, livable communities for generations to come. TPL has protected more than 3.65 million acres and completed 4,500 park, land and water conservation projects nationwide. In Minnesota, TPL has completed 158 project closings, permanently protecting nearly 100,000 acres of parks and land, including dozens in the Mississippi Headwaters.

To help public agencies acquire and restore land, build and improve parks, and fund park maintenance, TPL assists communities in securing public financing. TPL's conservation finance program offers technical assistance to elected officials, public agencies and community groups to design, pass and implement public funding measures that reflect popular priorities. Since 1996, TPL has been involved in over 570 successful ballot measures and dozens of statewide legislative campaigns creating more than $79 billion in new funding for parks, land conservation and restoration. Voters across the nation have approved 82 percent of the ballot measures directed and supported by The Trust for Public Land and TPL's lobbying affiliate, The Trust for Public Land Action Fund.

In Minnesota, The Trust for Public Land has been involved in ten (out of eleven) successful park conservation ballot measures, generating more than $5.8 billion for conservation most of which is dedicated to the Clean Water, Land and Heritage state constitutional amendment in 2008.
The Trust for Public Land is looking for new, creative ways to invest in nature-based solutions to manage climate change impacts in Minnesota with a focus on "natural climate solutions" to enhance nature's ability to absorb and store carbon in forests, wetlands, and agricultural soils. This new climate mitigation strategy requires prioritization by state and local decision makers.

Given the substantial investment of time and resources required for a successful conservation finance initiative, preliminary research is essential to determine which options and funding levels are feasible, economically prudent, and likely to be publicly acceptable. This research provides a stand-alone, fact-based reference document that can be used to evaluate financing mechanisms from an objective vantage point.[1]

[1] This report is based on the best information publicly available at the time of drafting: November 2018 to July 2019.

EXECUTIVE SUMMARY

The key to slowing climate change is to reduce carbon dioxide emissions, and the land can help: Natural areas like forests, farms, grasslands and wetlands, store and absorb carbon as they grow, and could do a lot more. U.S. forests remove between 12 and 14 percent of total U.S. greenhouse gas emissions each year through carbon sequestration, and offset nearly 16 percent of annual carbon dioxide emissions alone.[2] [3] More than 90 percent of land sector sequestration in the U.S. is attributed to forests, which capture carbon rapidly, in great quantity, and for long periods of time, storing it in tree trunks, leaves, branches, roots, and soil including peatlands. Wetlands in the United States store a total of 11.52 billion metric tons of carbon, much of which is within soils deeper than 30 cm. Wetlands in forested regions of the East and Upper Midwest store the most carbon, accounting for nearly half the wetland carbon in the United States.[4]

The potential for the land to offset a growing percentage of U.S. carbon emissions can be accomplished through conservation, restoration, and improved management of these natural resources. Reforestation has the largest mitigation potential to increase carbon absorption, followed by improved forest management practices.[5] Such actions not only mitigate climate impacts, but also provide additional co-benefits: resilient ecosystems reduce threats from disturbances such as wildfire, drought, and floods; while protecting fish and wildlife habitat, drinking water supplies, working farms and forests, and recreational opportunities.

The Trust for Public Land undertook this feasibility analysis to explore the state and local public funding options available in Minnesota to further natural climate solutions.

In order to understand what would be an appropriate funding source or sources, this report first provides a brief overview of the state's economy and budget. Next, the report provides an overview of existing programs that fund land conservation and other natural climate solutions and compares those programs to other states' programs. Then, the report analyzes possible revenue options for funding natural climate solutions, including, where available, estimates on revenue-raising capacity and examples of successful, similar programs in other states.

There are a number of potential public funding sources in Minnesota that can be woven together to further natural climate solutions in the Mississippi Headwaters.[6] Federal, local, and private conservation programs each have a role to play to help achieve these objectives. However, at the heart of the most successful conservation, restoration and improved management programs is a substantial, long-term, dedicated source of state revenue. With a consistent and reliable source of funds, Minnesota can establish and meet meaningful statewide climate goals and priorities that protect valuable natural resources, reflect community values, and leverage funds from federal agencies and private partners.

[2] Proceedings of the National Academy of Sciences Oct. 2017, www.pnas.org/content/114/44/11645
[3] http://www.ncsl.org/research/environment-and-natural-resources/state-forest-carbon-incentives-and-policies.aspx
[4] Nahlik, A.M. and M.S. Fennessy. 2016. Carbon Storage in U.S. Wetlands. Nature Communications 7:13835.
[5] (Griscom et al. 2017), https://www.pnas.org/content/114/44/11645
[6] https://global.nature.org/initiatives/natural-climate-solutions/ncs-about

Strategically conserving, restoring and managing natural resources is becoming increasingly urgent as Minnesota faces threats from climate change impacts. Several funding recommendations in this report can be considered by state and local governments and partners for natural climate solutions including: general obligation bonds, lottery revenue, the sales tax, use of existing sales tax on sporting goods, corporate franchise tax revenue, deed tax, mortgage registry tax, lodging tax, carbon tax, marijuana taxation, and new incentives for private landowners to promote natural climate solutions.

The next steps for Minnesota should include narrowing funding options to those that match the needs identified in the state's planning processes. Partners may also consider testing voter attitudes toward a specific set of funding proposals, which is essential for any proposal that requires voter approval on a state or local ballot. A public opinion survey of voters would test ballot language, tax tolerance, and program priorities of voters throughout the state.

This feasibility study is not a legal document and should not be relied upon for legal purposes.

THE MISSISSIPPI HEADWATERS[7]

The Mississippi Headwaters includes all or part of the following counties: Beltrami, Hubbard, Cass, Clearwater, Itasca, Aitkin, Mille Lacs, Isanti, Anoka, Crow Wing, Morrison, Todd, Wadena, Becker, Sherburne, Meeker, Wright, and McLeod.

The Upper Mississippi River (UMR) Basin, which contains the Headwaters, covers about 20,100 square miles, and stretches from the headwaters of the Mississippi River at Lake Itasca to Hastings, MN. Freshwater habitats in the Basin support 25 percent of all fish species in North America. More than 30 million people live and work in the UMR basin and rely on it for food, transportation, power production and drinking water.

The Headwaters is a largely forested watershed with lakes and wetlands located in the Northern Lakes and Forest ecoregion of Minnesota. The Headwaters watershed covers 1,255,105 acres in north-central Minnesota. Much of the runoff that flows from the Mississippi to the tributaries, lakes and groundwater passes through forested land which play a critical role in the overall high quality of water resources in this watershed because the forests serve as nature's purification system, filtering the water, absorbing pollutants and trapping sediment.

[7] Area is also considered the Upper Mississippi River Basin.

Forty-four percent of the land in the Mississippi Headwaters watershed is privately owned; the remaining land is owned by the state (36.1 percent), federal (18 percent), county (0.4 percent), and the Leech Lake Band of the Ojibwe Tribe (0.8 percent).[8]

The Ecological and Economic Significance of the Mississippi River Headwaters
The Mississippi River is the second longest river in North America, flowing 2,350 miles from its source at Lake Itasca to the Gulf of Mexico. (The Missouri River, a tributary of the Mississippi River, is about 100 miles longer.) What comes out of the Headwaters impacts the entirety of the Mississippi River.

Millions of people who live and work downstream rely on the Mississippi throughout a significant portion of the U.S., as one of the world's most important commercial waterways. There is more than $400 billion in annual economic impact for the whole basin and $345 million including one million jobs from the upper basin alone.[9] According to the National Park Service, barges and their tows move approximately 175 million tons of freight each year on the upper Mississippi alone.[10] The agribusiness industry that depends on the river produces 92 percent of the nation's agricultural exports, 78 percent of the world's exports in feed grains and soybeans, and most of the livestock produced nationally.[11] Along the length of the Mississippi, tourism, fishing, and recreation generate roughly $21.4 billion each year, and contribute 351,000 jobs.[12]

The Mississippi River serves as a major recreational resource for boaters, canoeists, hunters, anglers, and birdwatchers. In Minnesota, outdoor recreation generates 140,000 jobs and $1.4 billion in state and local tax revenue. A majority of these are associated with fishing and sightseeing by boat.[13] The River also serves as a critical source of drinking water for more than one million residents of Minnesota. Fourteen dams located in the headwaters power the generation of electricity for municipal and industrial uses throughout the watershed.[14]

The Mississippi Headwaters supports more than 350 species of animals, mammals and birds, including nearly all of the endangered, rare and threatened species listed in Minnesota. A dozen vegetative communities are found on the lands of the Mississippi Headwaters, representing nearly every type found in Minnesota.[15]

Protecting the Mississippi River Headwaters
Within Minnesota, the Mississippi River is protected from its Headwaters at Lake Itasca in Clearwater County, to the southern boundary of Morrison County by the Mississippi Headwaters Board (MHB). MHB is controlled by locally elected officials, with administration through county government and zoning authority provided by the Minnesota State Legislature, as well as the joint powers agreement of the eight counties surrounding the river. There is a total of 466 miles of river

[8] Sophia Vaughan. 2017. Groundwater Report –Mississippi River-Headwaters Watershed. *Minnesota Pollution Control Agency.* https://www.pca.state.mn.us/sites/default/files/wq-ws1-12.pdf
[9] Mississippi River Cities and Towns Initiative and www.umrba.org/umr-econ-profile.pdf
[10] National Park Service. "Mississippi River Facts". https://www.nps.gov/miss/riverfacts.htm
[11] Ibid.
[12] American Rivers. "Mississippi River: a cultural treasure". https://www.americanrivers.org/river/mississippi-river/
[13] Outdoor Industry Association. "Minnesota Fact Sheet". https://outdoorindustry.org/wp-content/uploads/2017/07/OIA_RecEcoState_MN.pdf
[14] Mississippi Headwaters Board. "Protection Efforts." http://mississippiheadwaters.org/history.asp
[15] Ibid.

that has been protected, the longest stretch of protected river in the U.S..[16] According to The Nature Conservancy, about 250,000 acres were converted to cropland between 2008 and 2012 with recent conversion particularly within the Mississippi Headwaters.[17]

Local partners and state agencies work together to monitor and assess the Mississippi River Headwaters watershed. State agencies include the Minnesota Pollution Control Agency, the Minnesota Department of Natural Resources and the Minnesota Board of Water and Soil Resources. Local partners involved include Hubbard Soil and Water Conservation District (SWCD), Itasca SWCD, Clearwater River Water District, Beltrami SWCD, Cass SWCD, Mississippi Headwaters Board, and the Leech Lake Band of Ojibwe.[18]

Leech Lake Band of Ojibwe

The Leech Lake Indian Reservation contains 864,158 acres of land, including parts of Beltrami, Cass, Hubbard, and Itasca Counties. It is located in North-Central Minnesota in the Central Pine-Hardwoods Forest ecoregion. Most of the Reservation lies within the Mississippi headwaters watershed, approximately 100,000 acres. The Mississippi River in the central part and the Boy River in the southeastern part drain it.

Mississippi Headwaters Board

The Mississippi Headwaters Board is an eight-county board created in 1980, with the signing of the Joint Powers Agreement. It was authorized by the Minnesota Legislature in 1981 to preserve and protect the first 400 miles of the Mississippi River.[19] The member counties are Aitkin, Beltrami, Cass, Clearwater, Crow Wing, Hubbard, Itasca, and Morrison.[20]

The basic tool for protection is a comprehensive local zoning ordinance establishing the uses and building dimensions for lands within a set corridor. The corridor is 500 feet to 1,000 feet on either side of the river, except where public, paved roads less than 500 feet or 1,000 feet are within corridor boundary.[21]

A comprehensive management plan set forth in 1981 sets guidelines for activities on public lands in the river corridor and for affected private lands. This management plan is achieved through cooperative agreements with the Leech Lake Tribal Council, the Minnesota Department of Natural Resources and the U.S. Forest Service-Chippewa National Forest. Management policies are provided for the retention and improvement of existing public lands along the river for fish and wildlife habitat and recreational use. Administration of the management policies takes place through the Mississippi Headwaters Board.

Every five years, the MHB develops an inventory of river resources and presents it to the Minnesota State Legislature. This inventory lists natural, cultural, scenic, scientific, and

[16] Mississippi Headwaters Board. "Regulatory Management: Comprehensive Management Plan." http://mississippiheadwaters.org/comprehensiveManagementPlan.asp
[17] https://www.nature.org/content/dam/tnc/nature/en/documents/Business_Case_for_Protecting_the_Mississippi_2019.pdf
[18] Minnesota Pollution Control Agency. "Mississippi River - Headwaters Watershed." https://www.pca.state.mn.us/water/tmdl/mississippi-river-headwaters-watershed
[19] MN Stat. 103F.361. Subd. (1) and (2). MN Stat. 103F.361-377, Appendix 2).
[20] Mississippi Headwaters Board. "Management Plan" http://mississippiheadwaters.org/files/regmanagement/ManagementPlan.pdf
[21] Mississippi Headwaters Board. "Protection Efforts." http://mississippiheadwaters.org/history.asp

recreational sites on the Mississippi Headwaters. It also contains an assessment of the river's health and risks to the river. This inventory is used to identify and establish management strategies for lands requiring greater protection than the zoning provides.[22]

The Nature Conservancy's Minnesota Headwaters Fund

The Nature Conservancy created the Minnesota Headwaters Fund, to secure investments from companies, foundations, and individuals to conserve key lands that filter and regulate water supply. The fund supports land and water conservation and restoration projects in targeted watersheds in the Upper Mississippi River basin in Minnesota, including easements, stream bank, and floodplain restoration, and other projects that prevent pollutants such as nitrates and sediment from entering key rivers and lakes.[23]

Threats to the Mississippi Headwaters

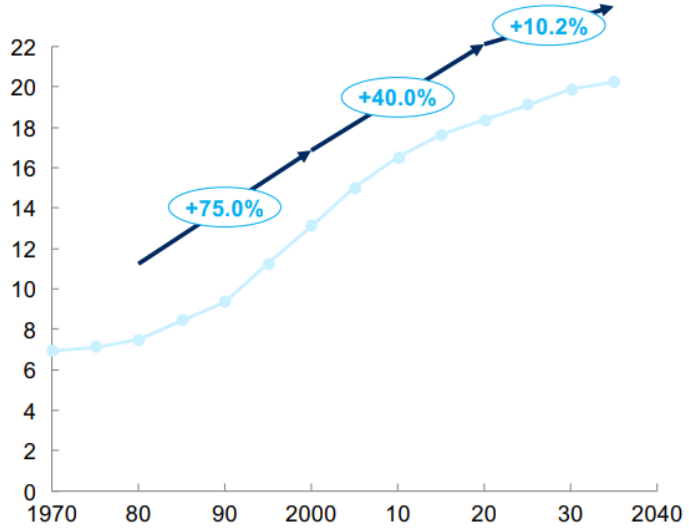
Most of the surface water resources within the Mississippi Headwaters watershed meet Minnesota's surface water quality standards. However, these resources are experiencing a loss of shoreline and aquatic habitat due to increases in population and development. Minnesota's population is likely to grow by approximately one million people by 2032. Much of that growth is expected to occur in the Twin Cities metro area and central Minnesota, where demand for Mississippi River water is already high.[24]

At the same time that demand for fresh water is increasing, rapid land conversion is threatening water quality. More than 260,000 acres of forest, wetland and grassland that drain into the Mississippi River above the Twin Cities were converted to agriculture and urban development between 2008 and 2013.[25] Additionally, there has additionally been increased nutrient, contaminant, and sedimentation from storm water runoff and forest management practices.[26] The following charts illustrated the continued threat to the Headwaters. A 2019 report by The Nature Conservancy and Ecolab introduces a 10-year, $400-600 million plan to restore and protect 200,000 acres of wetlands, grasslands and forests.[27]

[22] Mississippi Headwaters Board. "Regulatory Management: Comprehensive Management Plan." http://mississippiheadwaters.org/comprehensiveManagementPlan.asp
[23] The Nature Conservancy. "Minnesota Headwaters Fund Q&A." https://www.nature.org/en-us/about-us/where-we-work/united-states/minnesota/stories-in-minnesota/minnesota-headwaters-fund-qa-with-doug-shaw/
[24] Minnesota State Demographic Center. "Our Projections". https://mn.gov/admin/demography/data-by-topic/population-data/our-projections/
[25] Minnesota Pollution Control Agency. "Our Upper Mississippi River: Large River and Basin Restoration and Protection Strategies. " https://www.pca.state.mn.us/sites/default/files/wq-ws4-38b.pdf
[26] Minnesota Pollution Control Agency. "Mississippi River - Headwaters Watershed." https://www.pca.state.mn.us/water/tmdl/mississippi-river-headwaters-watershed
[27] Mississippi Headwaters: The Business Case for Conservation https://www.nature.org/content/dam/tnc/nature/en/documents/Business_Case_for_Protecting_the_Mississippi_2019.pdf

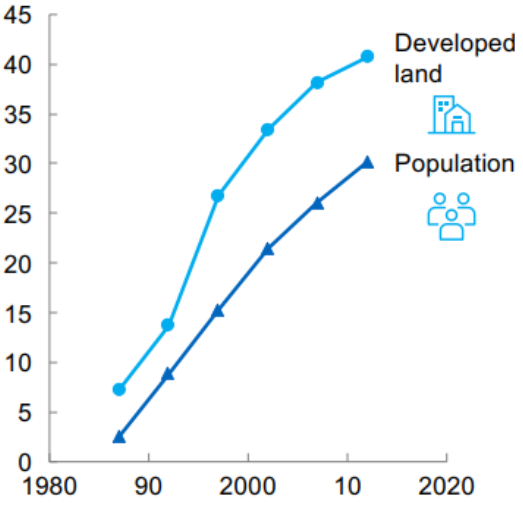
Average nitrogen loads in the Mississippi River Headwaters

Kilograms of nitrogen, millions



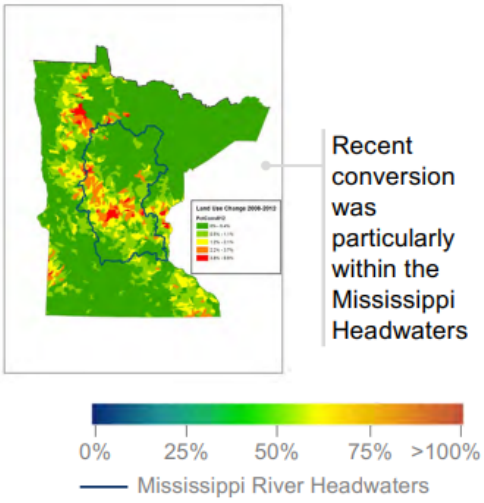
About 100,000 acres were converted for development from 2007 to 2012

Percent growth since 1982



About 250,000 acres were converted to cropland

Relative cropland expansion, 2008-2012[1]



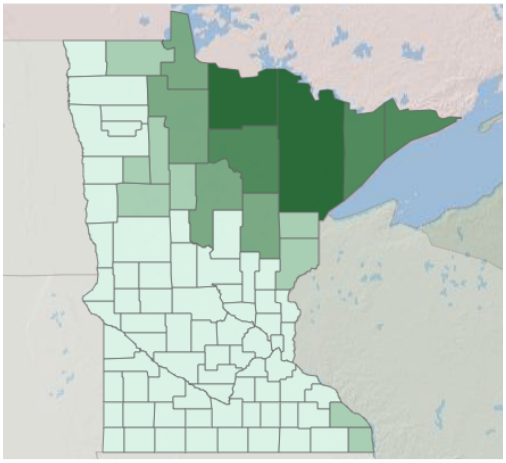
CLIMATE CHANGE ADAPTATION, RESILIENCE, AND MITIGATION

Minnesota is already feeling the effects of a changing climate, including extreme, unexpected, and fluctuating droughts, floods, and hot and cold spells. One of the tools the state can use to prepare for climate change impacts and mitigate carbon dioxide emissions are nature-based solutions. Nature-based solutions are strategies that conserve, create, restore and employ natural resources to enhance climate adaptation, resilience, and mitigation. They mimic natural processes or work in tandem with human-made engineering approaches to address natural hazards, like flooding, erosion, drought, and heat islands. Nature-based solutions can also help to maintain healthy natural cycles that sequester and maintain carbon and other greenhouse gases.

Minnesota is a member of the U.S. Climate Alliance, a bipartisan coalition of governors committed to reducing greenhouse gas emissions consistent with the goals of the Paris Agreement. The state has committed to 30 percent greenhouse gas reductions below 2005 levels by 2025 in the near term. In the long term their target is 80 percent greenhouse gas reductions below 2005 levels by 2025. The U.S. Climate Alliance Natural and Working Lands Initiative is focused on helping states use land-based resources to sequester and store carbon.

Carbon Sequestration

The natural areas throughout Minnesota sequester over 1.2 billion tons of carbon, making it the sixth largest state for volume of carbon storage.[28] Of the ten counties in Minnesota that sequester the most carbon, half are within the Mississippi River Headwaters: Aitkin, Beltrami, Cass, Clearwater, and Itasca. The 18 counties in the Mississippi Headwaters study area sequester 416 million tons of carbon or about 34 percent of the carbon stock for the state. The following maps created by The Trust for Public Land, depict the highest pocket of carbon storage in the state.



[28]The Trust for Public Land Carbon Map. https://dev.tplgis.org/carbonmap/

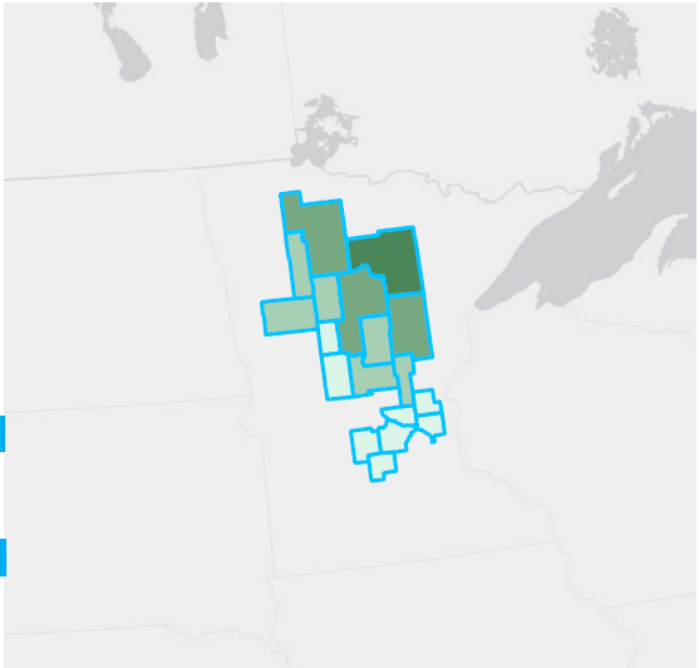
Forest Carbon Map

Mississippi Headwaters Summary Report

List of Minnesota Counties in Report

Hubbard, Becker, Aitkin, Beltrami, Wadena, Clearwater, Itasca, Crow Wing, Cass, Isanti, Sherburne, Wright, Anoka, Meeker, Todd, Morrison, Mille Lacs, McLeod

All of the calculations, data sources, and assumptions in this report are from EPA's Greenhouse Gas Equivalencies Calculator available at https://www.epa.gov/energy



Total Forest Carbon (All Stocks)		
416,584,383		
Metric tons		
Social Value	**Total Market Value**	
$18,346,376,207	$22,214,403,857	
U.S. dollars	U.S. dollars	

Gasoline Equivalence	Miles driven by avg. passenger vehicle	Homes' energy use	Pounds of coal burned	Acres of U.S. forests
172,033,834,127	347,469,246,338	165,104,178	1,672,718,472,524	1,798,664,334
gallons annually	miles annually	for one year	lbs annually	preserved from conversion to cropland

Top-5 Rank of Minnesota Counties by Total Forest Carbon Stocks

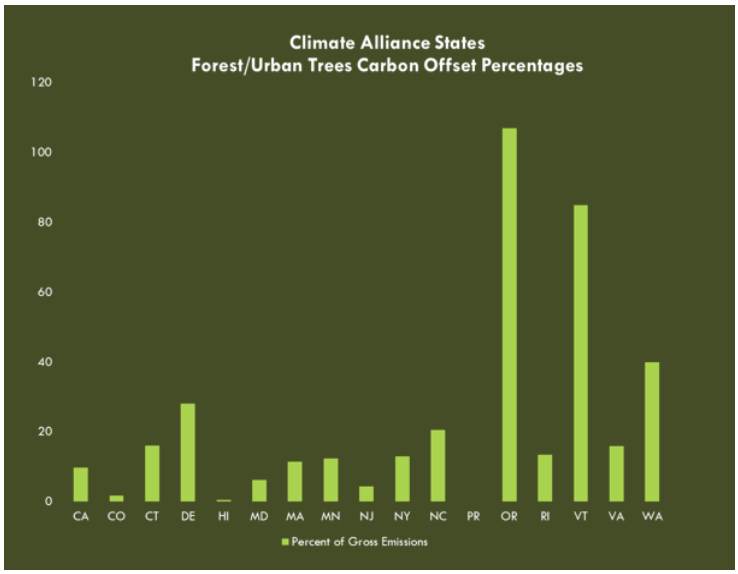
Rank	Name	Forest Carbon (metric tons)
1.	Itasca	105,096,284
2.	Cass	61,110,314
3.	Beltrami	60,644,038
4.	Aitkin	58,520,701
5.	Hubbard	26,740,820

The Trust for Public Land creates parks and protects land for people, ensuring healthy, livable communities for generations to come.The American Forest Foundation works on-the-ground with family forest owners, partners and elected officials to promote stewardship and protect our nation's forest heritage.

THE TRUST FOR PUBLIC LAND

AMERICAN FORESTS

Over the past two decades Minnesota has made great strides in sequestering and storing carbon emissions, however, Minnesota lags behind other U.S. Climate Alliance States in carbon offsets by forest and urban trees.[29] The following chart illustrate this situation.



NCS OPPORTUNITIES IN MINNESOTA

The natural climate solutions identified in this report will revolve around three topics (1) avoided conversion of specific land types, (2) forest management for increased carbon storage, and (3) agricultural management for increased carbon storage. Through a diverse set of funding options and policy proposals and changes, Minnesota can take steps to address the following solutions:

Avoided conversion of forests, farmland, grasslands, and wetlands

Conversion of certain types of land are a major source of the carbon release. For example, grasslands are being lost at a rate of over one million acres per year. When grassland is converted to cropland, about 28 percent of the carbon in the top meter of soil is released to the atmosphere. This trend could be reversed by re-enrolling 13 million acres of marginal cropland in conservation programs and restoring them to provide habitat and storage of carbon in the soil.[30]

Forest management for increased carbon

In a 2018 study by The Nature Conservancy that studied 21 natural climate solutions, increased reforestation (the planting of trees) emerged as the most effective means to achieve greater carbon storage, equivalent to eliminating the emissions of 65 million passenger cars. Other high-performing forest solutions include allowing longer periods between timber harvest to increase

[29] Several other states have since joined the U.S. Climate Alliance
[30] https://www.nature.org/en-us/explore/newsroom/natural-climate-solutions-study/

carbon storage; increasing controlled burns and strategic thinning in forests to reduce the risk of megafire; and avoided loss of forests from urban sprawl.[31]

Agricultural management for increased carbon

The loss of carbon from agricultural soils has been, in part, attributed to tillage, a common practice providing a number of benefits to farmers. The promotion of less intensive tillage practices and no tillage (the absence of mechanical soil disturbance) aims to mitigate negative impacts on soil quality and to preserve soil organic carbon (SOC).[32] Farmers can also plant cover crops, which sequester carbon out the atmosphere and return it to the soil during times of the year when fields would normally be bare.

CURRENT CONSERVATION FUNDING IN MINNESOTA

The most straightforward way to preserve, restore and expand the land-based carbon sink in the Mississippi River Headwaters, is to capitalize on existing government programs that support conservation, restoration, and improved management of natural systems. The key is to identify state programs that naturally align, or can be amended, to incentivize climate mitigation opportunities. To learn more about current statewide conservation programs and funding in Minnesota, please see the Appendix of this report.

FISCAL HEALTH, BUDGET AND DEBT

The Minnesota economy is doing extremely well. According to the State Economist, Minnesota has a 2.8 percent unemployment rate - the fifth lowest in the nation. According to the forecasts, the state is on track to have a surplus of $1.544 billion for fiscal years 2020 and 2021 (the entire state budget is expected to be $48 to $50 billion). The state's budget reserve, or "rainy-day fund" has grown to $2.075 billion.[33]

In July 2018, Standard and Poor's (S&P) upgraded Minnesota's credit rating to AAA, its highest rating. The affirmation of Minnesota's prime fiscal health follows a continued AAA rating from Fitch. In its determination, S&P touted the state's improved financial position and recently passed pension reform.

Minnesota's public pension system was reformed in 2018 to address $16.2 billion in unfunded liabilities. Which were subsequently reduced by $3.4 billion. The remainder will be paid down over time upon implementation of a number of reforms included in the measure. Approximately 511,000 public employees across the state now have a secure pension system to rely upon in their retirement. [34]

[31] Ibid.
[32] Environmental Evidence Journal Abstract: How does tillage intensity affect soil organic carbon? A systematic review, 2017
[33] Minnesota Has A Projected $1.5 Billion Budget Surplus — But What Does It Mean? By Dav Orrick, Pioneer Press December 6, 2018
[34] State of Minnesota Comprehensive Annual Financial Report 2018

State Debt Capacity Forecast[35]

As of November 30, 2018, the state of Minnesota had an outstanding principal totaling $7,808,565,000, which includes $6,305,475,000 in outstanding principal for general obligation bonds (both various purpose and trunk highway bonds) and $1,503,090,000 in outstanding principal for other tax-supported obligations. The state had no general obligation short-term notes outstanding.

Adherence to a debt management policy signals to rating agencies and the capital markets that a government is well managed and therefore is likely to meet its debt obligations in a timely manner.

The Capital Investment Guidelines are:

1. Total tax-supported principal outstanding shall be 3.25 percent or less of total state personal income.
2. Total amount of principal (both issued, and authorized but unissued) for state general obligations, state moral obligations, equipment capital leases, and real estate capital leases are not to exceed 6 percent of state personal income.
3. 40 percent of general obligation debt shall be due within five years and 70 percent within ten years, if consistent with the useful life of the financed assets and/or market conditions.

Guideline #1	
Tax-supported principal sold	$7.809 billion
FY 2019 state personal income estimate - IHS forecast	$324.508 billion
As a percent of state personal income, not to exceed 3.25%	2.41%
Estimate of max. amount of tax-supported debt that could be sold today before exceeding Guideline #1*	$2.738 billion*
**Bonds authorized are typically not sold all at once; they are sold over several years.*	
Guideline #2	
Total principal sold plus authorized and not yet sold	$12.380 billion
FY 2019 state personal income estimate - IHS forecast	$324.508 billion
As a percent of state personal income, not to exceed 6.0%	3.82%
Estimated max amount of additional tax-supported debt that could be authorized before exceeding Guideline #2	$7.090 billion
Guideline #3	
Of the state's general obligation bonds outstanding on June 30, 2018, 42.1 percent were scheduled to mature within five years and 73.3 percent were scheduled to mature with ten years. Furthermore, of the state's general obligation bonds expected to be outstanding on June 30, 2019, 42.1 percent are scheduled to mature within five years and 73.6 percent are scheduled to mature with ten years. The state is currently within Guideline #3.	

Maximum Debt Capacity under Current Guidelines

The maximum debt capacity refers to the amount of additional debt that could be authorized each legislative session without exceeding debt guidelines. The table below shows the maximum annual new debt authorizations until the limit in either Guideline #1 or Guideline #2 is reached. The calculations assume that the maximum amount of state debt would be authorized. The table also shows the additional general fund debt service costs resulting from the maximum authorizations.

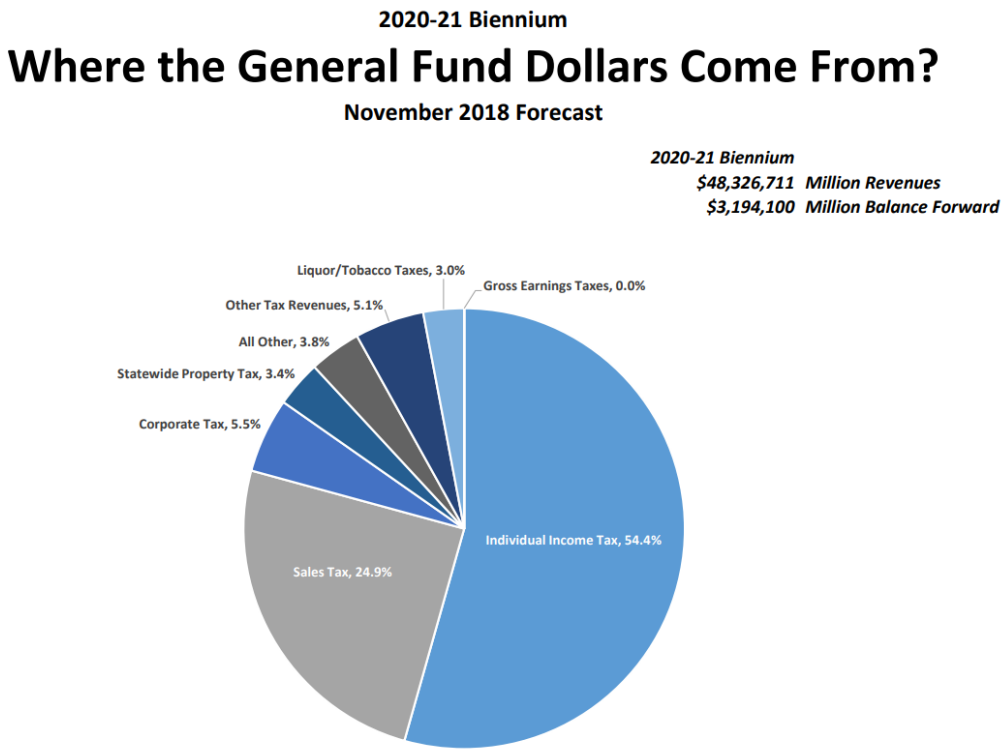
[35] Largely excerpted from "Debt Capacity Forecast," Minnesota Management and Budget, November 2018.

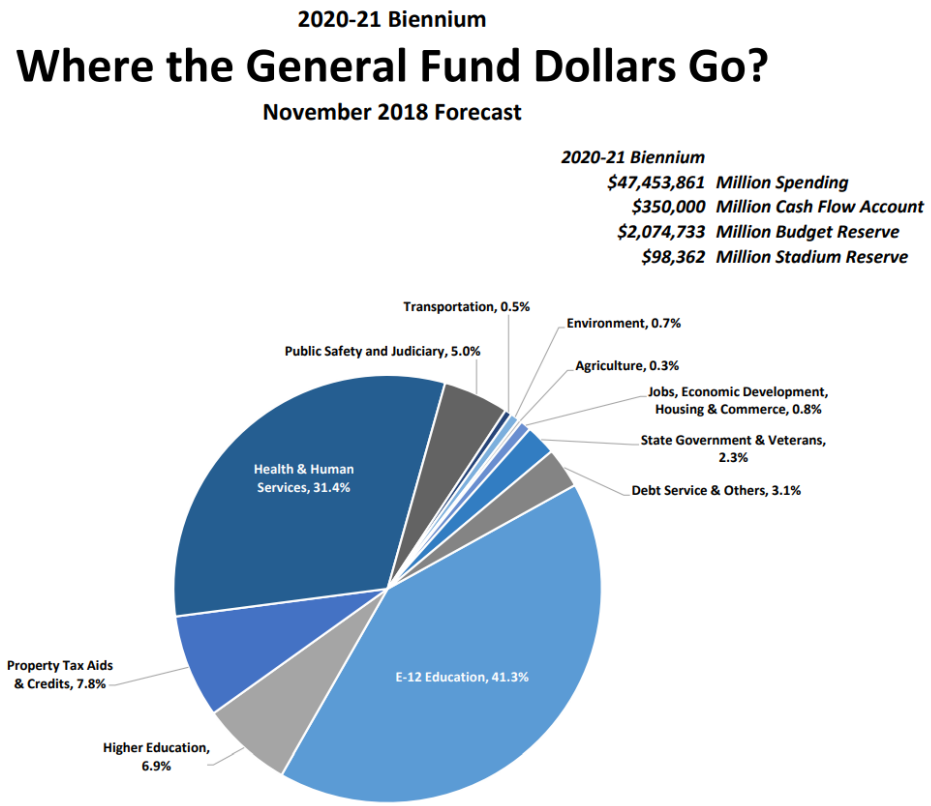
Maximum Annual New Debt Authorizations						
Dollars in millions	**FY 2019**	**FY 2020**	**FY 2021**	**FY 2022**	**FY 2023**	**FY 2024**
Bonding Bill Assumed in Nov '18 Forecast	$265	$755	$265	$755	$265	$755
Maximum New Debt Authorizations Within Debt Guidelines	$3,475	$1,550	$620	$1,725	$1,650	$1,300
Impact on Guideline #1	2.41%	2.57%	3.03%	3.25%	3.25%	3.24%
Impact on Guideline #2	3.82%	4.51%	4.55%	4.34%	4.41%	4.42%
Additional Debt Service Required	$0	$69	$265	$325	$369	$417

The amounts listed in the "Maximum New Debt Authorizations" line could be allocated among any of the following types of debt: various purpose general obligation bonds, trunk highway general obligation bonds, state general fund appropriation bonds, certificates of participation and real estate and equipment capital leases, and the assumptions for personal income are based on the November 2018 forecast.

The State Budget

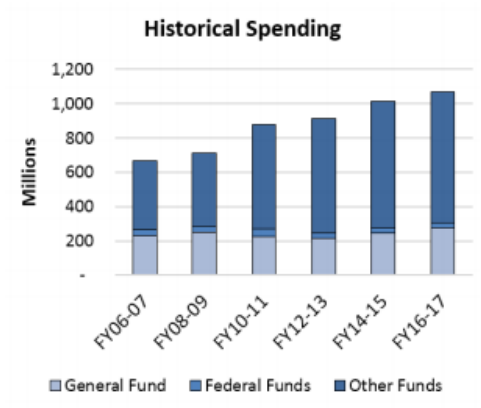
The following charts show a breakdown of the current general fund revenue and expenditures estimates for 2020 and 2021. Close to three quarters of the budget is made up of income and sales tax revenue. Only 0.7 percent is spent within the "environment" category.

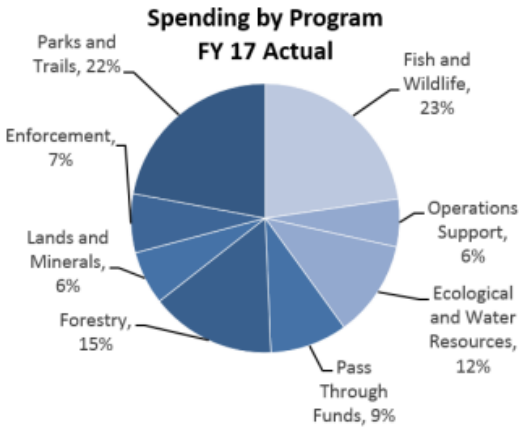




Minnesota Department of Natural Resources[36]

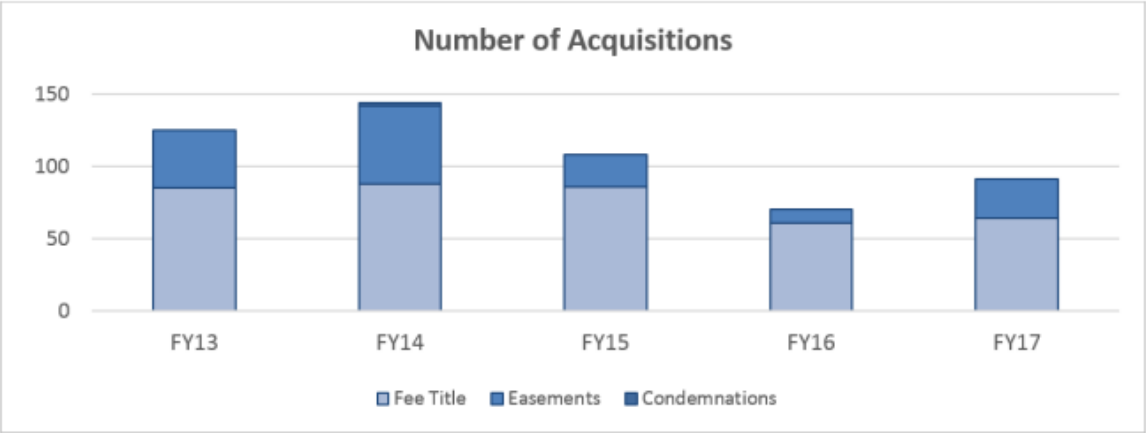
Over the past six years, spending by the Department of Natural Resources (DNR) has continued to increase at a steady pace. More than half of funding is spent on parks and trails (22 percent), fish and wildlife (23 percent), and forestry (15 percent). DNR manages 59 state forests covering 4.2 million acres for clean water, wildlife habitat, quality timber, outdoor recreation, and biological diversity.



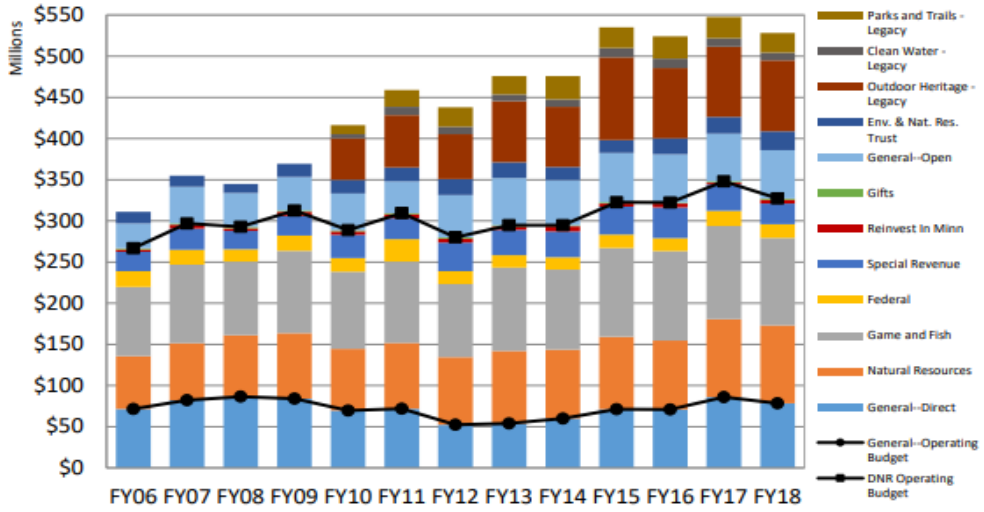


[36] Excerpted from 2018-19 DNR Biennial Budget

The Division of Lands and Minerals is responsible for negotiating land acquisition for the DNR. The graph shows the number of easement acquisitions increased in FY17 after a significant decrease the two previous fiscal years. Fee acquisitions are a more stable component, showing more modest variation through the years, although there has also been a decline in direct legislative appropriations to DNR for acquisitions due to dedicated funding from the sales tax and lottery proceeds.[37]



The DNR budget has changed over the past decade. Most notably, new revenues generated through the Clean Water, Land and Legacy Amendment has provided the opportunity to focus on special projects outside of regular base spending. The top line tracks DNR's operating budget, which has seen a slight increase over time (averaging two percent annually). The bottom line shows General Fund budget over the same period, which follows a similar trend.



[37] Personal communication with Minnesota Department of Natural Resources.

Tourism, Outdoor Recreation and the Economy

In addition to the natural goods and services provided by protected lands, public investment in land conservation in Minnesota contributes to a thriving outdoor recreation economy. According to the Outdoor Industry Association, 70 percent of Minnesota residents participate in outdoor recreation. Together, residents and nonresidents spend $16.7 billion each year on outdoor recreation, which generates $1.4 billion in state and local tax revenue. Such spending also supports 140,000 jobs and $4.5 billion in wages and salaries. In fact, more jobs in Minnesota directly depend on outdoor recreation than hospital and health systems.[38]

Lands protected by the state of Minnesota also support viable populations of fish, game, and other wildlife species. These lands are accessible for wildlife watching, fishing, and hunting, which are critical to the state's economy and culture. In fact, Minnesota ranks tenth in the nation for the number of resident hunters. In 2011, there were nearly 457,000 resident hunters who spent $670 million on hunting related retail sales. In the same year, there were 477,000 total hunters across the state, and these hunters spent a combined 5.6 million days hunting.[39] The economic contribution of all hunting activity across the state was $733 million in retail sales, which had a total multiplier effect of $1.3 billion and supported over 12,000 jobs with $418 million in salaries and wages.[40] Looking beyond hunting, there were 1.7 million anglers and 1.6 million wildlife-watching participants in Minnesota as well. These anglers and wildlife-watchers spent $2.4 billion and $621 million on trips and equipment, respectively. Considered together, there were 2.5 million wildlife related recreationalists who spent a total of $3.9 billion on wildlife related recreation expenditures in Minnesota.[41]

[38] Outdoor Industry Association, Minnesota, https://outdoorindustry.org/wp-content/uploads/2017/07/OIA_RecEcoState_IL.pdf
[39] This includes resident and non-residents.
[40] Southwick, *Hunting in America*, https://www.fs.fed.us/biology/resources/pubs/wildlife/HuntingEconomicImpacts-NSSF-Southwick.pdf
[41] U.S. Fish and Wildlife Service, U.S. Census Bureau, *2011 National Survey of Fishing, Hunting, and Wildlife-Associated Recreation*. While 2016 survey results are available at the national level, the 2016 National Survey of Fishing, Hunting, and Wildlife-Associated Recreation did not collect data at the state level, so there are no state-level reports. Thus, 2011 is the most recent year for which data are available.

BEST PRACTICES FROM OTHER STATES

Substantial State Investment

The foundation to implement effective natural climate solutions is strong fiscal commitment on the part of state government through stable revenue sources. Substantial and direct state investment fosters program development and long-term vision. Some existing state programs rely on a single revenue stream, while others use a combination of dedicated revenue sources.

Commonly utilized revenue sources by states for land conservation funding are general obligation bonds, general fund appropriations, lottery proceeds, sales taxes, real estate transfer taxes and deed-recording fees. Other state revenue sources for conservation include license plate revenues, hunting and fishing license fees, hotel/motel taxes, cigarette taxes, state income taxes and oil and gas revenues. The highest level of security for conservation funding comes from constitutional dedication.

Common Revenue Streams for State Conservation Programs	
Revenue Stream	Selected Examples
General obligation bonds	California, Maine, Ohio
Budget appropriations	Arizona, Utah, Montana, North Carolina
Lottery proceeds	Colorado, Minnesota, Oregon
Sales taxes	Arkansas, Minnesota, Missouri, New Jersey
Real estate transfer taxes	Florida, Maryland, Delaware, Hawaii, Washington
Deed recording fee	New Hampshire, Massachusetts, Connecticut
Source: The Trust for Public Land's Conservation Almanac	

Since 1996, 55 (of 64) statewide conservation finance and park measures have been approved around the country and are often relied upon to protect natural resources under threat of conversion or development. Many state conservation funding programs are created with voter approval and others through the legislative process.

For example, in June 2018, California voters approved a $4.1 billion bond with 58 percent support. The bond is dedicated to conservation, park and climate programs; and funds state and local parks, environmental protection and restoration projects, water infrastructure and flood protection projects. This is one of the few funding measures that has explicitly dedicated funds to the effects of climate change.

Midwestern U.S. Funding Case Examples

These funding examples below are statewide programs focused on land conservation. These existing funds have the potential to be used for land conservation projects to help implement natural climate solutions.

Michigan: In 1984, voters passed a constitutional amendment establishing the Michigan Natural Resources Trust Fund and provided constitutional protection for the acquisition of land for public recreation and environmentally significant lands. Financial support is provided to local

governments and the Department of Natural Resources through oil and gas lease revenues. About $30 million is generated annually for conservation purposes. Such funds could potentially be used to help implement a state natural climate solutions plan.

Missouri: In 1976, Missouri voters passed a permanent 1/8 of one-cent sales tax, which generates $90 million to $100 million a year for management of Missouri's fish, forestry and wildlife. These funds are administered by the Department of Conservation. The state also has a separate, dedicated sales tax for state parks, soil and water conservation. In 1984, Missouri voters approved a constitutional amendment that implemented a one-tenth of one percent sales tax. The tax generates approximately $90 million annually and revenue is split equally between these two programs. These funds are administered by the Department of Natural Resources. The tax provides about 75 percent of annual funding for the state parks department.[42] Voters reauthorized the tax in 1988, 1996, and 2016. At least two-thirds of voters approved the reauthorization each time it appeared on the ballot.[43]

Nebraska: In 1992, Nebraska voters approved a constitutional amendment that created the Nebraska Lottery and the Environmental Trust. Of the total revenues, 44.5 percent are to be used for environmental protection purposes. The remaining funds are used for education purposes. In FY16, the Trust Fund received $18.8 million in lottery proceeds. The lottery has transferred more than $260 million to the Trust in the last 22 years.[44] Trust Fund revenue currently supports grants in the following priority areas: habitat, surface and ground water, waste management, air quality, and soil management.[45]

Ohio: In 2000 and 2008, voters approved constitutional amendments that secured $800 million in bond funding for the Clean Ohio Fund. The Fund restores, protects, and connects Ohio's important natural and urban places by preserving green space and farmland, improving outdoor recreation, and cleaning up brownfields to encourage redevelopment and revitalize communities.

[42] Sherman, Lucille. "Voters asked to renew parks and conservation tax for the fifth time." Columbia Missourian. October 12, 2016. Accessed June 27, 2017. http://www.columbiamissourian.com/news/state_news/voters-asked-to-renew-parks-and-conservation-tax-for-the/article_c12a63f6-8c95-11e6-b2e1-c350aae73d9f.html
[43] "Parks, Soils and Water Sales Tax." Missouri State Parks. Accessed June 27, 2017. https://mostateparks.com/page/55069/parks-soils-and-water-sales-tax
[44] "2016 Annual Report." Nebraska Environmental Trust." Accessed June 20, 2017. http://www.environmentaltrust.org/publications/annual%20reports/2016_Annual_Report.pdf
[45] "Our Priorities." Nebraska Environmental Trust. Accessed June 20, 2017. http://environmentaltrust.org/about/our_priorities.html

CHOOSING A FUNDING STRATEGY

As noted above there are several types of broad-based revenue sources available to state governments to pay for natural climate solutions: three primary funding mechanisms include discretionary spending (i.e. budget appropriations), dedicated funding streams, and general obligation bonds. The financing options utilized by a state will depend on a variety of factors such as taxing capacity, budgetary resources, voter and legislator preferences, and political will. While many states can create funding for conservation through their budgetary process, this either happens infrequently or does not yield adequate funding.

In The Trust for Public Land's experience, whether for conservation funding, or funding for specific projects to promote natural climate solutions, governments that create funding via the budget process generally create less funding than those that create funding through voter supported ballot measures. As elected officials go through the process of making critical budgetary decisions, funding for parks and land conservation sometimes lags behind other public purposes and behind what voters usually support.

With their own funding for natural climate solutions, state governments are better positioned to leverage funding with federal agencies and private philanthropic partners funding from federal agencies and private philanthropic partners. Having predictable funding sources empowers the state to establish long-term climate priorities that conserve, restore and manage forests and wetlands to help communities become resilient to climate change impacts. In addition, sustainable funding will help absorb, store and mitigate carbon dioxide emissions while creating co-benefits, such as the conservation of habitat, water supplies, working farms and forests, parks and trails.

When evaluating the range of funding options available for natural climate solutions in Minnesota, it is helpful to keep several factors in mind. These funding favorability factors include:

- **Nexus between funding source and natural climate solution:** Is there a natural linkage between the funding source (like a sporting goods sales tax dedication) and climate mitigation, resilience and mitigation?
- **Revenue raising capacity:** Will the revenue source raise enough money to accomplish reasonable conservation objectives? Establishing a new funding source requires significant effort and must generate adequate funding to warrant the effort.
- **Reliability:** Will the source provide consistent funding, or will it suffer fluctuations due to the economy? Will it be subject to diversions to other emerging priorities?
- **Competition:** Are there other state agencies or programs competing for the same funding source?
- **Precedent:** Has another state been successful in using the funding option? Breaking new ground can be more challenging when there is no precedent.
- **Political feasibility:** Is there a political champion who will lead the effort?
- **Lack of organized opposition:** Another aspect to political feasibility is the presence of likely, paid opposition. Certain funding options could have predictable opponents that can be hard to neutralize.

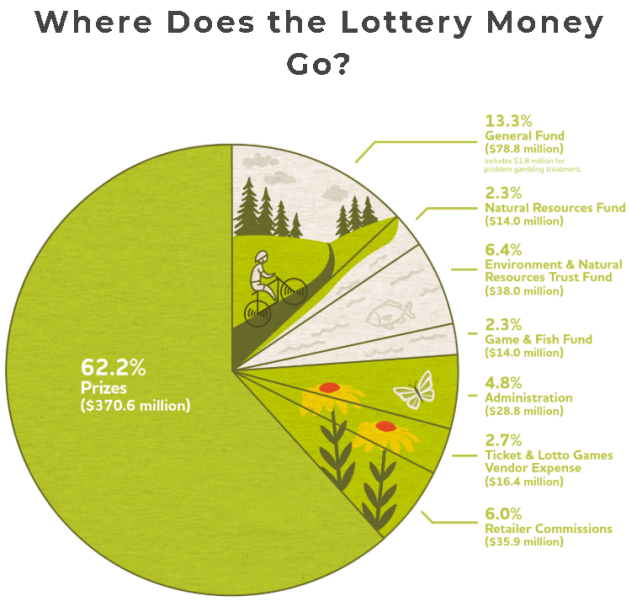
- **Potential partners:** Can supporters of natural climate solutions align themselves with other groups to mutually establish a new funding source (e.g., the arts, sportsmen, water resource protection, recreation, outdoor industry, etc.)?

STATE REVENUE OPTIONS FOR CONSERVATION

This Feasibility Report focuses on potential revenue options, incentives, and regulations that are significant, pragmatic, and proven. This study examines general obligation bonds, lottery revenue, the sales tax, use of existing sales tax on sporting goods, corporate franchise tax revenue, deed tax, mortgage registry tax, lodging tax, carbon tax, marijuana taxation, and new incentives for private landowners to promote natural climate solutions. Most taxes, fees, and regulations discussed in this section are statutory and to amend them, the house and senate must pass a bill to increase the rates and the governor must sign the bill. The alternative to this would be a constitutional amendment dedicating revenue similar to the lottery and sales tax.

Lottery

In November 1988, Minnesota voters passed a constitutional amendment (with 81 percent support) that permanently established the Environment and Natural Resources Trust Fund (Trust Fund).[46] The Trust Fund supports projects with the public purpose of protection, conservation, preservation, and enhancement of the state's air, water land, fish, wildlife, and other natural resources. The Trust Fund receives support through lottery proceeds for land conservation. In November 1990, voters (again with 81 percent support) approved dedication of 40 percent of net lottery proceeds to the Trust Fund through December 2000. In 1998, voters (77 percent support) passed another constitutional amendment extending the dedication through December 2024. Funds may also be generated through private donations. Once deposited into the Trust Fund, contributions become part of the balance and are invested in a combination of stocks and bonds. The income generated from these investments is reinvested back into the Trust Fund.



Trust Fund proceeds are appropriated to individual projects by the Minnesota Legislature, based upon recommendations of the Legislative-Citizen Commission on Minnesota Resources (LCCMR). LCCMR members review, evaluate, and rank all submitted proposals and the Commission then

[46] Minnesota Constitution Art. XI, Sec.14

decides upon a selection of proposals to recommend for funding the lottery is an important source of funds for conservation, it currently returns about $38 million[47] per year to the Trust Fund out of $600 million in total FY18 annual receipts. **In FY18, approximately $77 million in undesignated lottery funds went to the state's general fund.**

In addition to Environment and Natural Resources Trust Fund, the lottery also funds DNR's Natural Resources Fund and Game and Fish Fund.

The need for additional funding is evident as there is an annual gap between the amount of funding requested and the amount of funding approved. In July 2018 the LCCMR had recommended 67 projects totaling $53,696,000. However, the LCCMR received 273 proposals totaling $191 million through a competitive, multi-stage evaluation. For the FY20 and FY21 biennium (July 1, 2019 - June 30, 2021), approximately $61 million is available each year for the Environment and Natural Resources Trust Fund. (Most of the approximately $500 million and 1,000 projects approved since the start of the program have been south of the Mississippi Headwaters, not invested in the study area of this report.)

In addition to renewing the portion of the lottery dedicated to the Trust Fund before it expires at the end of 2024, dedicating additional funding from the portion of the lottery set for the general fund could be another option to boost funding for natural climate solutions.

The Minnesota Constitution requires that "not less than 40 percent" of the net proceeds of the state lottery must be credited to the Environment and Natural Resources Trust Fund. There would be no constitutional barrier to depositing more than 40 percent to the trust fund, or to dedicating additional funds for a specific purpose, such as natural climate solutions. Either of these could be accomplished by a change in statute. Additionally, the amount of the constitutional dedication could be changed by a ballot measure dedicating a greater portion of lottery revenues to the Trust Fund.

Even taking the same 40 percent from the total amount deposited to the state instead of the "net proceeds" would have generated over $20 million more for the Trust Fund. In addition, there are different ways the lottery proceeds could be used including dedicating a certain portion of funds to be used for the benefit of natural climate solutions.

Lottery Revenue for Conservation in Other States

Colorado

In 2018, the Colorado legislature reauthorized the Colorado Lottery, which last year generated $135 million for parks, trails, playgrounds, open space and conservation, through the year 2049. Lottery proceeds are distributed through the Great Outdoors Colorado Trust Fund ($63.7 million last year), Colorado Parks and Wildlife ($14.3 million last year) and the local Conservation Trust Fund (CTF) ($57 million last year). CTF funds are distributed on a per capita basis to towns,

[47] Trust Fund funding only applies to the "net proceeds" ($95M in FY18)

counties, cities and special districts for parks and recreation. Based upon 2017 numbers, a 25-year extension could potentially generate $3.375 billion.

Oregon
In 1998, voters approved a ballot measure that dedicates 15 percent of net lottery proceeds to a Parks and Natural Resources Fund. Revenue allocated to the Fund is divided equally between state parks and watershed enhancement (e.g., floodplain enhancement). This source provides substantial funding for land conservation, restoration, and state parks projects.[48] In FY17 it generated over $180 million for these purposes. Prior to the passage of this measure, all lottery proceeds benefited K-12 schools and economic development. In November 2010, Oregon voters approved Measure 76 with 69 percent support, which permanently extended the lottery dedications first approved in 1998, as that authorization was set to expire at the end of 2014.

Measure 76 was citizen-initiated: close to 200,000 signatures were submitted to the secretary of state. Lottery money also spurred a public-private effort to restore streams, wetlands and critical habitats for salmon, other fish and other wildlife.

Nebraska
In 1992, Nebraska voters approved a constitutional amendment that created the Nebraska Lottery and the Environmental Trust. 44.5 percent of lottery revenues are to be used for environmental protection purposes. In FY17, the Trust Fund received $18.1 million in lottery proceeds. The lottery has transferred more than $280 million to the Trust in the last 24 years.[49] Trust Fund revenue currently supports grants for habitat, surface and ground water, waste management, air quality, and soil management.[50]

Dedicating a Portion of Existing Sales Tax Revenue

Minnesota could consider dedicating the portion of existing tax revenues from the sales of sporting goods to natural climate solutions. Texas, Virginia, and Georgia dedicate the portion of general sales tax revenue attributable to the purchases of sporting goods to parks and land conservation, which can include natural climate solutions. The Minnesota's Department of Revenues defines a category of retail sales for "sporting goods, hobby, music, and bookstores."

Minnesota state sales tax in this category in 2016 (the last year data was available) totaled roughly $80 million.[51] The state legislature could dedicate the entire amount, or a percentage, of these tax revenues each year for natural climate solutions.

Such a dedication would take away revenues currently allocated to other purposes and potentially necessitate cuts to other areas of the Minnesota state budget or create the need to generate additional funds from other sources. However, $80 million in sales tax revenue represents just 1.3 percent of total state sales tax revenue in FY18.

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[48] 2009 ORS 541.377
[49] "2017 Annual Report." Nebraska Environmental Trust." Accessed September 2018. http://www.environmentaltrust.org/publications/annual%20reports/2017%20Annual%20Report.pdf
[50] "Our Priorities." Nebraska Environmental Trust. Accessed June 20, 2017. http://environmentaltrust.org/about/our_priorities.html
[51] It is important to note that 0.375% of the state sales tax in Minnesota is constitutionally dedicated to various natural resources and cultural heritage funds, until June 30, 2034. The remaining 6.5% goes to the General Fund.

In addition to the constitutional Legacy sales tax dedication, three other statutorily created funds exist for sales tax revenue.

Revenue from legal firework sales are dedicated as follows:

- 25 percent to the Volunteer Fire Assistance Grant Account in the Special Revenue Fund
- 25 percent to the Fire Safety Account
- The remainder to the General Fund

Revenue from the 9.2 percent rental tax and the 5 percent rental fee are dedicated to the Highway User Tax Distribution Fund.

For the motor vehicle parts transfer, each month $12,137,000 is transferred from the General Fund to the Highway User Tax Distribution Fund.

Funds generated from sales tax on sporting goods in Minnesota could support an $800 million revenue bond with four percent interest rate over 20 years, and could be used strategically for climate adaptation, resilience, and mitigation.

Minnesota Revenue Bond Financing Estimates			
Bond Issue	Interest Rate	Maturity (Years)	Annual Debt Service
$200,000,000	4.0%	20	$14,716,350
$400,000,000	4.0%	20	$29,432,700
$600,000,000	4.0%	20	$44,149,050
$800,000,000	4.0%	20	$58,865,400

The state has occasionally used revenue bonds to support programs for public safety and retirement. [52]

Minnesota 2018 HF4437

There is some precedent in Minnesota for this type of dedication of existing revenue. In 2018, a proposed constitutional amendment in Minnesota would have asked voters to dedicate sales tax revenues generated from auto parts and repairs toward fixing roads, bridges, highways and other transportation projects. The House passed the measure easily, but it did not come to a vote in the Senate. The measure would have allocated $300 million annually.[53]

In addition, in 2006, 60 percent of voters chose to constitutionally dedicate the motor vehicle sales tax to highways and public transportation.

Georgia Outdoor Stewardship Act

In 2018, Georgia lawmakers referred a constitutional amendment to the November ballot that would create a permanent funding source for land and water conservation. The Georgia Outdoor

[52] https://mn.gov/mmb/debt-management/bonding/bond-sale/revenue-bonds/archived-revenue-bonds.jsp
[53] https://www.revisor.mn.gov/bills/bill.php?b=house&f=HF4437&ssn=0&y=2018

Stewardship Act was overwhelmingly approved by voters with 83 percent support. The new state constitutional amendment will dedicate a portion of the existing sales and use tax on outdoor sporting goods to land conservation and parks for ten years. Funding will protect forests (including working forests), wildlife habitat, and land critical to clean drinking water, wetlands, and to acquire and improve parks and trails in communities throughout the state.

Texas
In November 2019, after years of having sporting goods sales tax revenue reallocated from parks, the Texas state legislature and Governor approved Proposition 5 for the November 2019 ballot. Proposition 5 added language to the Texas Constitution to dedicate revenue from the sales tax on sporting goods to the state Parks and Wildlife Department and the state Historical Commission. Whereas the former law allowed the legislature to allocate the revenue for other uses, Proposition 5 would require a two-thirds vote in each legislative chamber to reduce the amount for the parks, wildlife, and historical agencies. Furthermore, the ballot measure would prohibit the legislature from decreasing the amount for the parks, wildlife, and historical agencies by more than 50 percent.[54] The measure passed with 88 percent support and will generate over $3.5 billion over 20 years.

Mortgage Registry Tax and Deed Tax[55]

Minnesota's mortgage registry tax (MRT) and deed taxes are two separate state taxes that apply to many real estate transactions. The MRT is based on the amount of debt secured by a mortgage of real property and is imposed when the mortgage is recorded. The deed tax is a transfer tax; it is imposed on the value of real property transferred. While these taxes are independent of each other, they are often thought of together, since many property transactions trigger both taxes.

The MRT rate is 0.23 percent of the total debt. The deed tax rate is 0.33 percent of the price paid for the real property. These rates have been in effect since 1987. County treasurers collect these taxes. They remit 97 percent of the revenues to the state for deposit in the state general fund. The county retains the other 3 percent for its administrative expenses. Collections are sensitive to the volume and value of real estate transactions; MRT collections are also sensitive to refinancing activity. Following the impacts of the 2007-2009 recession, historically low interest rates and a slowly improving economy have spurred steady growth in collections for the past several years.

Hennepin and Ramsey Counties have special provisions that allow them to impose an additional mortgage and deed tax for brownfields remediation.[56]

The state collected $122 million in FY18 from the MRT in FY18. As an example, .07 percent increase in the tax would generate over $37 million annually.

The state collected $128 million from the deed tax in FY18. As an example, .07 percent increase in the tax would generate over $27 million annually.

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[54] Excerpted from https://ballotpedia.org/Texas_Proposition_5,_Sales_Tax_on_Sporting_Goods_Dedicated_to_Parks,_Wildlife,_and_Historical_Agencies_Amendment_(2019)
[55] Excerpted from Mortgage and Deed Taxes: An Overview Minnesota House Research October 2016
[56] https://www.revisor.mn.gov/statutes/cite/383A.80; https://www.revisor.mn.gov/statutes/cite/383B.80

In January 2019, Massachusetts Governor Baker proposed a new source of funding for climate resiliency projects (which include natural climate solutions) to be funded by an increase in the state's real estate transfer tax. Currently, the state imposes a tax on any transfer of real estate ownership, equal to $4.56 per $1,000 of property value. The transfer tax is levied on the seller. Governor Baker's recommendation would raise that rate to $6.84 per $1,000 of property value. According to the Baker Administration, this would raise between $130 million and $150 million per year that would be allocated toward a state-managed fund to address climate change.

In July 2019, the Massachusetts House of Representatives unanimously approved "GreenWorks" legislation. Modeled after the state's MassWorks program, GreenWorks is a $1.29 billion bond bill aimed at making significant investments in resiliency, clean energy, and energy efficiency. The legislation establishes a 10-year grant program to fund clean energy, energy efficiency, and climate change resiliency measures that cut greenhouse gas emissions, fortify infrastructure and reduce municipal costs.

Real estate transfer taxes in other states

Several states use a real estate transfer tax to fund parks and land conservation, including Illinois, Pennsylvania, and Tennessee. In November 2014, voters in Florida approved a constitutional amendment for a 20-year dedication of 1/3rd of existing revenue generated by the documentary stamp tax (same as a transfer tax) to fund open space, parks, recreation, farmland, watershed protection, trails, greenways, and forests. The measure was expected to raise $18 billion and passed with 75 percent support.

There is a reasonable nexus between real estate development and land conservation, and Minnesota could consider increasing the current fee and dedicating the revenue to such purposes. However, since revenues from the tax fluctuate with the real estate market, income can be difficult to predict. In addition, the creation of a new tax or increase of an existing fee involving real property is usually strongly opposed by the real estate industry – a well-funded opponent.

Lodging Tax

Most states tax lodging at the local level, however over half of the states also impose a tax or fee at the state level. Minnesota does not. Over $158 million was generated through the state sales tax for lodging purposes in FY16, the most recent data available. This money goes to the state general fund.

Minnesota law prohibits local governments from imposing a sales tax without legislative authorization. A local government may impose a tax on lodging in one of two ways:

- The local government receives authorization under a special law.
- A city or town may impose a tax of up to three percent on lodging under general authorization provided in Minnesota Statutes, Section 469.190. Of the total proceeds from

the tax, 95 percent must be used to fund a local convention or tourism bureau.[57] Currently there are about 120 jurisdictions in Minnesota that levy a local lodging tax. These taxes generated close to $77 million according to 2017 figures. [58]

A recently enacted lodging tax in Plymouth (Laws 2019, 1st. spec. sess. Ch. 6, art 6, sec. 25) is the first special lodging tax where a portion of the revenue raised is specifically dedicated to capital improvements of public recreational facilities. Most local jurisdictions administer and collect their own local lodging taxes and the assumption is that they use the remaining five percent to cover those collection costs, but nothing in the statute requires them to use the five percent for a specific purpose. Jurisdictions could follow the Plymouth model above and receive legislative approval for a lodging tax with funding dedicated to natural climate solutions. One drawback is that the Hospitality Industry of Minnesota lobbies heavily against these diversions.

The chart shows local governments that impose a lodging tax. Note that in Minnesota, a tax exclusively on out-of-state visitors would be unconstitutional.[59] Article X states, "Taxes shall be uniform upon the same class of subjects…".

	Rate	Year Imposed
Special Laws:		
Biwabik (Giants Ridge)	2.0%	2011
Bloomington	7.0%	1970
Brooklyn Center	3.0%	2003
Cook County	1.0%	2008
Duluth		
More than 30 Rooms	5.5%	1970
Other	3.0%	1970
Minneapolis		
More than 50 Rooms	5.125%	1986
Other	3.0%	1969
Proctor	3.0%	2006
Rochester	7.0%	1971
St. Cloud	5.0%	1979
St. Paul		
More than 50 Rooms	6.0%	1986
Other	3.0%	1970
Two Harbors	1.0%	1994
Winona	1.0%	1991

General Authorization: about 120 cities and towns impose this tax.

Revenue Collections:		
	C.Y. 2016	$75,393,000
	C.Y. 2017	$76,544,000

30 states tax lodging statewide from a range of $5 per room (Georgia) to 15 percent (Connecticut). If the state imposed a one percent lodging tax for example, it could generate over $23 million annually for natural climate solutions.[60]

Corporate Franchise/Income Tax[61] [62]

A flat tax rate of 9.8 percent applies to Minnesota taxable corporate income as defined under subchapter "C" of the Internal Revenue Code. The tax was first implemented in 1933 and its current rate has held steady for nearly 20 years.

FY18 actual revenues were $1.31 billion or about 6 percent of general fund revenues. Revenues are deposited in the general fund. Revenues under the tax are the most volatile of the major state taxes. When the economy goes into recession, corporate profits and the franchise tax revenues can drop

[57] Minnesota Tax Handbook: A Profile of State and Local Taxes in Minnesota 2018 Edition
[58] https://www.house.leg.state.mn.us/hrd/pubs/lodgetax.pdf
[59] Note that in Minnesota, a tax exclusively on out-of-state visitors would be unconstitutional.
[60] Based on state sales tax statistics provided by the Minnesota Department of Revenue
[61] Corporate Franchise Tax, MN House Research Short Subjects, June 2015
[62] Minnesota Tax Handbook: A Profile of State and Local Taxes in Minnesota 2018 Edition

quite precipitously. For example, in FY07 (an expansion year), revenues were $1.17 billion; by fiscal year 2010 (a recession year) they had dropped to $663 million (a reduction of 43 percent reduction from 2007), and by fiscal year 2012 they had recovered to $1.04 billion (an increase of 57 percent from 2010).

Potential revenue using a one-half percent example could generate over $67 million annually in Minnesota (based on total collections in FY18) for land and water conservation, as well as natural climate solutions.

Corporate Franchise Tax example: New Jersey
In November 2014, 65 percent of New Jersey voters approved of a constitutional amendment dedicating $2.15 billion over the next 20 years for various state land conservation programs including Green Acres, using a portion of corporate business taxes collected by the state. Prior to 2014, New Jersey voters had approved 13 consecutive bonds for land conservation dating back to 1961. The measure increases the percentage of the corporate business tax for environmental programs to six percent by 2019. This was not a tax increase. Instead, it reallocated funds that were previously destined for water programs, underground storage tanks, and hazardous site cleanups and dedicated them to open space, farmland, and historic preservation. Green Acres received $105 million for open space preservation in first round of awards in 2017.

Comparison of New Jersey Corporate Business Tax			
Previous Allocation		Allocation after 2014 Amendment	
Percentage	Purpose	Percentage	Purpose
15%	Water Quality	71%	Preservation of open space, farmland, historic sites, and flood prone areas
25%	Hazardous substance discharge remediation (brownfields) and underground storage tank removal and cleanup	10%	Hazardous Substance discharge remediation (brownfields)
28%	Polluted site cleanup performed by the State	9%	underground storage tank removal and cleanup
17%	Diesel air pollution control (This allocation expired after 12/31/15 and was to be realllocated to improvements in parks and other preserved open space)	5%	Polluted site cleanup performed by the state
15%	Improvements in parks and other preserved open space	5%	Water quality
On July 1, 2019, the amount dedicated to these programs will increase from four to six percent. The allocations will also shift so that the preservation of open space, farmland, historic sites, and flood prone areas will increase to 78 percent of dedicated funds.			
In FY14, this four percent generated $103.1 million.			
Source: League of Women Voters of New Jersey			

Recreational Marijuana

A growing number of states, including Minnesota, allow or are considering recreational marijuana and accompanying taxes on its sale. There are no states that currently dedicate marijuana tax revenue to parks and conservation specifically, although California dedicates a portion of its revenue to environmental programs through the Departments of Fish and Wildlife and Parks and Recreation to clean up and prevent damage from illegal marijuana cultivation. This revenue source is included in this report due to its potential to generate substantial amounts of revenue, although revenue is usually dedicated to public safety and prevention efforts.

Marijuana in Minnesota

Voters in November 2018 made Michigan the first Midwestern state (and 10th overall) to legalize recreational marijuana.[63] At least three bills have been introduced in Minnesota; one would put the questions to voters in November 2020 via constitutional amendment.[64] The bills do not discuss at what rate recreational marijuana will be taxed, but they do set aside future revenue: for police training to recognize drivers impaired by the psychoactive component of marijuana; for mental health treatment; and for public education on the effects of the drug — especially among children.[65]

Recreational Marijuana tax examples

Michigan

In November 2018, 56 percent of voters in Michigan approver Proposal 1, a constitutional amendment legalizing recreational marijuana. The measure created an excise sales tax of 10 percent, which was to be levied on marijuana sales at retailers. Proposal 1 required that revenue from the tax be dedicated to local governments, K-12 education, and road and bridge maintenance. The ballot initiative also legalized the cultivation, processing, distribution, and sale of industrial hemp. Municipalities were authorized to ban or limit marijuana establishments within their boundaries.[66] The Department must first spend the money collected for implementation, administration, and enforcement, and then, for at least two years, $20 million each year to one or more clinical trials researching the efficacy of marijuana in treating veterans and preventing veteran suicide. The unspent balance is allocated as follows: [67]

- 15 percent to municipalities in which a marijuana retail store or microbusiness was located, allocated in proportion to the number of those establishments within the municipality.
- 15 percent to counties in which a marijuana retail store or microbusiness was located, allocated in proportion to the number of those establishments within the county.
- 35 percent to the School Aid Fund for K-12 education.
- 35 percent to the Michigan Transportation Fund for repair and maintenance of roads and bridges.

Once the market matures, Michigan is expected to generate $134 million annually from recreational marijuana. This does not include the revenue from the state six percent sales tax.

California

In November 2016, California voters approved Proposition 64, which legalized personal marijuana use and implemented a state tax on marijuana sales. As of January 2018, marijuana sales will include a 15 percent excise tax. The California Legislative Analyst's office stated that the amount of excise tax revenue is uncertain due to many economic and regulatory factors.[68] As such, the office estimated that additional tax revenues could range from "high hundreds of millions of dollars to over $1 billion annually" and could reduce criminal justice costs by tens of millions of dollars

[63] "After marijuana's legalization elsewhere, why not Minnesota? By Steve Karnowski Star Tribune January 28, 2019
[64] Minnesota House File 265. House File 420 and Senate File 619 are also being considered.
[65] Is the Legalization of Recreational Marijuana in Minnesota Inevitable? Twin Cities Business, January 31, 2019 Peter Callaghan
[66] https://ballotpedia.org/Michigan_Proposal_1,_Marijuana_Legalization_Initiative_(2018)
[67] Michigan Senate Fiscal Agency Analysis of Proposal 1
[68] Proposition 64 Revenues." California Legislative Analyst's Office. February 16, 2017. Accessed June 20, 2017. http://www.lao.ca.gov/handouts/crimjust/2017/Proposition-64-Revenues-021617.pdf

annually.[69] The most recent quarter of data came out in May 2019, providing five quarters in all. The total over the last four quarters was $214 million - far below the projections initially made due to initial slow growth in the regulated market.[70]

The measure makes specific allowances for state programs totaling $65 million and allocates remaining revenue to various purposes, including 20 percent to clean up and prevent environmental damages resulting from the illegal growing of marijuana. Revenue is placed into an Environmental Restoration and Protection Account managed by the Department of Fish and Wildlife and Department of Parks and Recreation.[71] A majority vote of the legislature could modify the programs funded with a majority vote beginning in 2028-29.

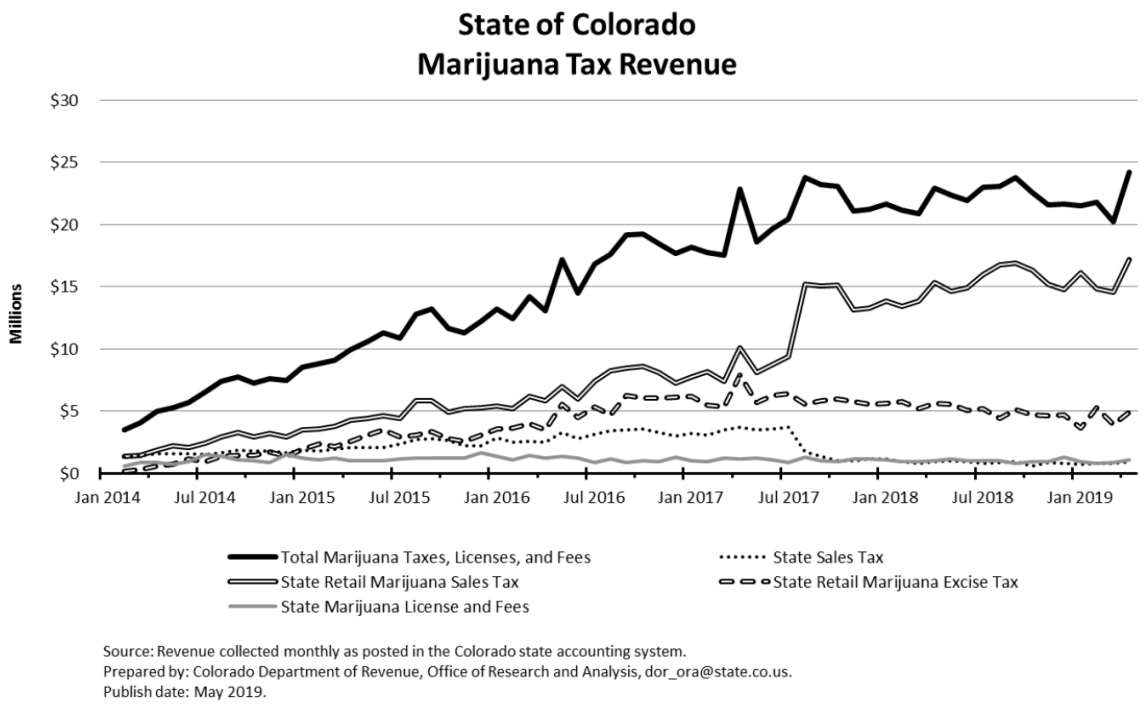
Colorado

Colorado legalized medical marijuana in 2000 and recreational marijuana in 2014. Starting July 1, 2017, the state retail marijuana sales tax rate is 15 percent, the state share is 90 percent of the total tax collected, and the local government share is 10 percent of the total tax collected.

For FY17-18, the first $40 million of the Retail Marijuana Excise Tax revenue was distributed to the Public School Capital Construction Assistance Fund (PSCCAF) administered by the Colorado Department of Education's Building Excellent Schools Today (BEST) program. Excise tax collections in excess of $40 million, $27.8 million for FY17-18, were transferred to the Public School Fund.

The chart below demonstrates the growing marijuana tax revenue, which totaled over $266 million in 2018.

[69] "Proposition 64." California Legislative Analyst's Office. November 8, 2016. Accessed June 20, 2017. http://www.lao.ca.gov/BallotAnalysis/Proposition?number=64&year=2016
[70] Personal communication with California's Legislative Analysis Office.
[71] "Proposition 64 Revenues." California Legislative Analyst's Office. February 16, 2017. Accessed June 20, 2017. http://www.lao.ca.gov/handouts/crimjust/2017/Proposition-64-Revenues-021617.pdf



Sports Betting Authorization and Taxation[72]

The U.S. Supreme Court cleared the way for states to allow bets on sports in its May 2018 ruling in *Murphy v. NCAA,* which repealed the federal Professional and Amateur Sports Protection Act of 1992. That law had prohibited states from "authorizing" gambling related to professional and amateur sports leagues.

As of the date of this writing, Arkansas, Delaware, Indiana, Iowa, Mississippi, Montana, Nevada, New Jersey, New Mexico, New York, Oregon, Pennsylvania, Rhode Island, and West Virginia authorize sports betting in some form, although sports betting operations are not yet up and running in some of those states.[73] Several other states have passed bills that are waiting for final approval.

In 2018, nineteen states proposed legislation to legalize sports betting. Delaware was the first state to take bets, opening its doors to the activity in June 2018. Delaware's law allows the state to keep 50 percent of the proceeds from the activity. Mississippi and West Virginia began offering bets in August. New York, Pennsylvania, and Rhode Island have legalized sports betting, but have not begun taking bets.

[72] "Legalized, Taxed Sports Betting Brings Millions to States," Bloomberg Tax, September 6, 2018. https://www.bna.com/legalized-taxed-sports-n73014482814/ (Accessed November 14, 2018).
[73] "Sports betting in Minnesota" House Research" October, 2018

Some estimates say $2 billion could be wagered annually in Minnesota, and a one percent tax is currently being discussed via legislation. There could be opposition from anti-gambling groups and getting support from local tribes would be important.[74]

If a sports betting tax of one percent were levied on this revenue in Minnesota, it could generate $20 million annually for natural climate solutions.

In March 2019, a bill to legalize sports betting in Minnesota cleared its first test with the approval of the Senate Tax Committee despite overwhelming tribal opposition. The bill would allow sports betting only at the state's two horse-racing tracks, Canterbury Park in Shakopee and Running Aces in Columbus; its 21 tribal casinos; or via mobile device apps linked directly to the tracks or participating casinos. The state would collect a 6.75 percent tax on total bets placed minus winnings.[75] The Minnesota Department of Revenue said the tax impact of the bill was as yet "unknown."

In November 2019, voters in Colorado approved Proposition DD, which legalized sports betting in Colorado, with most proceeds dedicated to the state's Water Plan. The ballot measure read as follows:

Shall state taxes be increased by twenty-nine million dollars annually to fund state water projects and commitments and to pay for the regulation of sports betting through licensed casinos by authorizing a tax on sports betting of ten percent of net sports betting proceeds, and to impose the tax on persons licensed to conduct sports betting?

Proposition DD authorizes the state to collect up to $29 million per year in sports betting tax revenue. Tax revenue from sports betting is expected to vary as the market matures. Based on information from other states' sports betting markets, about $16 million in tax revenue is expected to be collected on average over the first five years. Close to 95 percent of the revenue will be dedicated to water projects and other water-related obligations that address the state's water needs including projects funded under the Colorado Water Plan and water obligations outlined in interstate water compacts and federal law.[76]

Impervious Surface Fees

States and municipalities may choose to implement stormwater utility fees for pollution abatement, stormwater runoff management, water quality protection, habitat protection, and natural climate solutions. As of June 2018, there are 1,681 documented stormwater utilities within the United States.[77] These utilities vary significantly in terms of geographic and population size, payment structure, and management structure.

[74] "Could legalized sports betting be coming to Minnesota sooner rather than later?" by Peter Callagan December 11, 2018
[75] https://minnesota.cbslocal.com/2019/03/07/bill-to-legalize-sports-betting-in-minnesota-clears-1st-hurdle/
[76] https://leg.colorado.gov/sites/default/files/initiative%2520referendum_2019-2020%20hb%2019-1327v3.pdf

[77] Campbell, C. Warren, Dymond, Randel, Kea, Kandace, Dritschel, Amanda (2018). "The Western Kentucky University Stormwater Utility Survey 2018," Stormwater Utility Surveys, Bowling Green, Kentucky.

In recent decades, stormwater agencies have increasingly turned to stormwater fees, rather than property taxes alone, to meet some stormwater management costs. Authority for a local or regional agency to enact and administer stormwater programs and assess user fees is most commonly derived from an enabling statute enacted by the state legislature or via the state's constitution or charter. Stormwater fees are in effect in 40 states and more than 1,600 jurisdictions nationwide. A stormwater fee typically extends to properties that are exempt from ad valorem property taxes, such as nonprofit and public properties, allowing the cost burden of stormwater management to be shared more broadly among the beneficiaries of the stormwater system.

Stormwater fees can take a variety of forms. Basic approaches include a Per-Parcel Fee, or a rate based on a parcel's total area such as a Per-Arce Fee. While fees under these approaches are easy to calculate, they do a poor job of linking a property's runoff and burden to the amount of the fee. At the other end of the spectrum are complex approaches that levy a pro-rated fee based upon several factors such as a measurement of the impervious area of land.

A basic consideration in implementing a stormwater fee is determining whom to bill. According to a 2014 study of stormwater utilities,[78] 71 percent of stormwater fees were charged to utility customers and appeared on monthly utility bills, while 29 percent were charged to property owners and appeared on property tax bills or separate bills.

Equivalent Residential Unit. The most widely used method of funding is the ERU system --Equivalent Residential Unit (ERU) is usually the average impervious area on a single-family residential parcel, although some communities define it as the average of all residential parcels. The ERU method is a hybrid of the Flat Per-Parcel Fee and the Total Impervious Area methods.[79] Under this method, stormwater utilities determine the average impervious area for a single-family residence in the service area. The average impervious area is designated as one "ERU" and assigned a rate. All residential properties are generally charged the rate of one ERU, with each paying the same rate regardless of size. Other property types, such as commercial, industrial and institutional, are charged based on actual measurements of impervious area, using ERUs as the unit of measurement.[80]

Minnesota Statutes 444.075 – Waterworks Systems; Storm, Sanitary Sewer Systems
Subd. 3b. Storm sewer charges.
Storm sewer charges may be fixed:
(1) by reference to the square footage of the property charged, adjusted for a reasonable calculation of the storm water runoff; or
(2) by reference to a reasonable classification of the types of premises to which service is furnished; or
(3) by reference to the quantity, pollution qualities, and difficulty of disposal of storm water runoff produced; or
(4) on any other equitable basis, including any combination of equitable bases referred to in clauses (1) to (3), but specifically excluding use of the basis referred to in subdivision 3a, clause (1); and otherwise without limit.

Many jurisdictions categorize fees by land use. In newer communities where zoning has produced uniform development patterns, it is a common practice to group single-family homes together and charge a single fee to the entire class. Nationwide, the average monthly single-family residential fee was $5.34, and the median fee was $4.00 per month. Most fees go up over time reflecting an increase in the Consumer Price Index (CPI).

[78] Campbell, C. Warren, Dymond, Randel, Kea, Kandace, Dritschel, Amanda (2014). "The Western Kentucky University Stormwater Utility Survey 2014," Stormwater Utility Surveys, Bowling Green, Kentucky.
[79] *Total Impervious Area*. Under this method, a stormwater fee is calculated based on a property's impervious area alone.
[80] Beneath the Surface: A Primer on Stormwater Fees in New Orleans. Bureau of Government Research, February 2017.

Minnesota

There are 198 stormwater utilities in Minnesota.[81] Monthly fees range from $1.00 in Maple Lake to $45 in Luverne. The range of fee amounts likely reflects stormwater needs and local political realities. According to the 2019 Western Kentucky Stormwater Utility Survey, over $120 million was generated through local stormwater utility fees. The following page shows a snapshot of jurisdictions in Minnesota with stormwater utility fees. Funding typically goes into an enterprise fund that can only be used for stormwater purposes (operations, maintenance, replacement, and construction of the storm sewer system such as pipes, manholes, catch basins. These are generally spelled out in the ordinance creating the stormwater utility.

The stormwater utility fee could be used in jurisdictions to fund natural climate solutions. In Minneapolis, the fee funds flood control and water quality improvement facilities, such as ponds, underground detention vaults, sump catch basins, or constructed wetlands. Green infrastructure such as bioretention basins, rain gardens, or tree trenches would be classified as water quality treatment facilities and would be funded by the stormwater utility fee.[82]

No.	Community	State	Fee Type	ERU (ft^2)	Monthly Fee	Year	Population	Ann. Rev.
782	Mankato	MN	A		$4.76		39,528	$1,269,232
783	Mantorville	MN	V				1,971	$16,435
784	Maple Lake	MN	D		$1.15		2,088	$13,807
785	Maple Plain	MN	T		$2.97	2005	1,792	$98,470
786	Mapleton	MN	V				1,756	$25,693
787	Maplewood	MN	R		$7.48	2003	38,472	
788	Marshall	MN	R			2003	13,700	$977,357
789	Mayer	MN	D		$4.00	2005	1,780	$18,857
790	Medina	MN	R		$2.53	2008	4,963	$205,150
791	Mendota Heights	MN	A		$2.42	1992	11,168	$276,602
792	Minneapolis	MN	E	1,530	$13.09	2005	387,753	$39,038,000
793	Minnetonka	MN	A		$7.08	2003	50,435	$2,026,316
794	Minnetonka Beach	MN	F		$5.00	2011	539	
795	Minnetrista	MN	F		$8.67	2004	6,474	$99,089
796	Montrose	MN	F		$4.50	2000	2,887	$43,980
797	Moorhead	MN	F		$10.21	2005	38,566	
798	Mora	MN	F		$1.25	2005	3,556	
799	Mound	MN	R		$1.00	2001	9,180	
800	Mounds View	MN	T			1993	12,305	
801	New Brighton	MN	R		$4.58	1994	21,715	
802	New Hope	MN	D		$7.96	1991	20,616	
803	New Prague	MN	R		$15.53	1992	7,401	
804	Newport	MN	R		$4.08		3,481	
805	North Branch	MN	R			2008	10,131	
806	North Mankato	MN	A		$3.75		13,437	
807	North Saint Paul	MN	T		$10.56	1990	11,601	
808	Northfield	MN	R			1986	20,084	
809	Norwood Young America	MN	R			2003	3,611	
810	Oak Park Heights	MN	T			1999	4,389	

Code	Meaning
IA	Impervious Area
E	ERU
F	Fixed Rate
T	Tier System
R	Residential Equivalence Factor (or similar)
D	Two Level System (Residential/Commercial)
X	No information
S	Fee per Parcel Square Footage
A	Fee per Parcel Acre
-	Repealed
M	Water Meter
W	By Water Usage
V	Existence of SWU/ Fee verified
AV	Ad valorem fee
LU	Land use based fee

[81] Authority provided under Minnesota Statutes § 444.075.
[82] Personal communication with the City of Minneapolis Public Works

Carbon Pricing

A carbon price or tax creates a price for emitting carbon dioxide (and potentially other greenhouse gases) to the atmosphere. Such a tax would support the Minnesota's other policies designed to meet the goals of the Paris Agreement. These goals require the state to reduce its greenhouse gas (GHG) emissions to 30 percent below the 2005 level by 2020 and to at least 80 percent below 2005 by 2050. In 2017, Minnesota became one of the initial signatories to the U.S. Climate Alliance to uphold the Paris climate deal commitment to reduce emissions by at least 26 percent from 2005 levels and meet the federal Clean Power Plan targets.

No one in recent years[83] has evaluated revenue from a potential carbon tax in Minnesota. One way to estimate potential carbon tax revenues in Minnesota, would be to use the Minnesota Pollution Control Agency's GHG emissions inventory[84] and multiply which emissions would be subjected to the tax by the tax amount (per ton of emissions). For example, the PCA estimates there was over 2.2 million metric tons of carbon released through the use of oil through agriculture, forestry, and land use practices in Minnesota in 2016 (the most recent year data is available).

In Minnesota, a $30 per metric ton of carbon tax on just this source could generate $66 million for natural climate solutions.[85]

Past Carbon Pricing Legislation

Companion Carbon Pricing bills were introduced in the Minnesota House and Senate in the spring of 2018. SF 4086 and HF 4517 did not make it through committee hearings. Both bills propose a fee would start at $40 per ton of CO2 in 2020 and increase by $5 for first five years. Beginning in the sixth year, it would increase by $10. Beginning in the 12th year, the fee would increase by $15. The bill is revenue neutral and would return the funds collected in the form of a rebate to individuals on a per capita basis. No official economic or revenue analysis had been completed for his legislation.

Carbon Pricing example: Washington[86]

So far, no states have passed a carbon tax. Ballot Initiative 732 in 2016 and 1631 in 2018 would have made Washington the first U.S. state with a carbon tax. According to the latest data published in 2018, Washington ranked 26th in carbon emissions nationwide (Minnesota ranks 24th).[87]

Initiative 1631 would have enacted a carbon emissions fee on large emitters based on the carbon content of fossil fuels sold or used in the state and electricity generated in or imported for use in the state. The fee would have been $15 per metric ton of carbon beginning on January 1, 2020 and

[83] In December 1991, the Institute for Local Self-Reliance proposed a $6 per ton carbon tax, in line with those imposed by European govenments at the time. Assuming a 27 million ton overall carbon emission level, and subtracting for industrial and agricultural use, a carbon tax at this level was estimated to generate about $150 million a year. Assuming an average cost of saved carbon of $100 a ton, this investment, if fully applied to energy efficiency, would reduce carbon emissions by 1.5 million tons a year, a little over 5 percent of l988 emissions. At around the same time, The Department of Natural Resources had recommended a carbon tax of 55 cents per ton dedicated to tree planting. This would generate $13.5 million a year. https://ilsr.org/wp-content/uploads/files/images/polluterreport.pdf
[84] https://www.pca.state.mn.us/air/greenhouse-gas-emissions-data
[85] Personal communication with Minnesota Pollution Control Agency
[86] Excerpted from Ballotpedia.org https://ballotpedia.org/Washington_Initiative_1631,_Carbon_Emissions_Fee_Measure_(2018)
[87] https://www.eia.gov/environment/emissions/state/analysis/pdf/stateanalysis.pdf

would have increased by $2 per metric ton each year until the state's greenhouse gas reduction goals of 2035 were met and the 2050 goals were on track to be met. Revenue from the fee would have gone into three funds: (1) for air quality and energy programs and projects, (2) for water quality and forest health projects, and (3) for investments related to communities. Under Washington law, this measure was a fee and *not* a tax because the revenue could not have been spent on government expenses or public programs; rather, it would have been dedicated to specific accounts related to investing in climate and environmental projects. This measure would have raised almost $2.3 billion in its first five fiscal years. It failed at the ballot with 57 percent voting against.

Specific to natural climate solutions, the measure would have dedicated 25 percent of revenue to the clean water and healthy forests account for investments related to water and forests such as programs, activities, or projects that:

- Restore and protect fisheries and marine habitats;
- Reduce flood risks and prepare for a rise in sea level; and
- Increase water supply

A clean water and healthy forests panel would have been created and tasked with providing detailed recommendations to the public oversight board regarding investments and other matters related to water and forest programs.

The ballot language was as follows:

Initiative Measure No. 1631 concerns pollution. This measure would charge pollution fees on sources of greenhouse gas pollutants and use the revenue to reduce pollution, promote clean energy, and address climate impacts, under oversight of a public board. Should this measure be enacted into law?

Carbon Pricing example: Boulder, Colorado[88]

In 2006, Boulder, Colorado passed the country's first carbon tax, in the form of the Climate Action Plan (CAP) Tax. The CAP tax is levied on city residents and businesses, based on the amount of electricity they consume; and the utility Xcel Energy charges consumers in their monthly utility bills. Consumers receive rate deductions for using electricity from renewable sources, particularly for utilizing Xcel's Wind Source program. Rates vary depending on the sector and cost an average family about $21 a year. In total, the CAP tax generates about $1.8 million in annual revenue.[89] Proceeds are directed to the city's Office of Environmental Affairs to fund programs that reduce greenhouse gas emissions, such as weatherization efforts, sustainability projects, and solar rebates. Profits could also go towards conservation, restoration, land management actions and other natural climate solutions.

States that Have Introduced Carbon Pricing Legislation

In January 2018, state-level lawmakers in nine states — Connecticut, Maryland, Massachusetts, New Hampshire, New York, Oregon, Rhode Island, Vermont, and Washington— announced the

[88] "Boulders climate action plan (CAP). Boulder public health and safety climate.
[89] https://www-static.bouldercolorado.gov/docs/CAP_document_2017_updated_FINAL-1-201709121536.pdf?_ga=2.123802120.38399057.1517344873-1460474048.1517344873

formation of a Carbon Costs Coalition that seeks to tax carbon.[90] Through this coalition, each state hopes to create and pass independent carbon pricing legislation, while utilizing the others for resources and ideas. Carbon bills have been introduced and re-introduced in over a dozen state legislatures including Maryland, Washington, New York, Hawaii, Rhode Island, Vermont, Massachusetts and Maine, and proposals to study a carbon tax have been introduced in New Mexico, New York, New Hampshire and Vermont.[91]

Cap and Trade

Cap-and-trade programs are government-mandated, market-based systems that set a limit on the total amount of greenhouse gas emissions allowed from various industries. This cap is divided into allowances and distributed to companies within the relevant industries. Companies that do not use all of their allowances can sell the remainder or save allowances for future use. This ability to sell emission allowances provides companies with an incentive to lower their emissions and invest in cleaner forms of energy. The cap progressively decreases, decreasing emission levels accordingly. Cap-and-trade programs have boasted high compliance rates and are an economically effective approach to reducing air pollution.[92]

The Regional Greenhouse Gas Initiative (RGGI) is the first mandatory market-based cap and trade program in the U.S. to reduce greenhouse gas emissions. RGGI is a cooperative effort among the states of Connecticut, Delaware, Maine, Maryland, Massachusetts, New Hampshire, New York, Rhode Island, and Vermont to cap and reduce Carbon Dioxide (CO2) emissions from the power sector. New Jersey is hoping to rejoin in 2019. In 2014, RGGI states implemented a cap of 91 million tons of CO2, which declines 2.5 percent each year from 2015 to 2020. In August 2017, RGGI states announced proposed program improvements, which would result in an overall 30 percent reduction in the regional cap from 2020 to 2030. The RGGI states have already reduced power sector carbon emissions by nearly 50 percent since 2008, while generating more than $2.7 billion in proceeds across the region for reinvestment in programs to benefit consumers and build a stronger and cleaner energy system.

In September 2018, an analysis was completed looking at the benefits of 2016 RGGI investments for all states in the pact. Those results are shown below.
[93]

[90] https://www.utilitydive.com/news/lawmakers-from-9-states-vow-to-put-a-price-on-carbon/516154/
[91] https://www.forbes.com/sites/patrickgleason/2018/02/26/state-lawmakers-compete-to-levy-nations-first-carbon-tax/#2aa670c83ac3
[92] Forest Climate Working Group: Tapping into U.S. Forests to Mitigate Climate Change http://forestclimateworkinggroup.org/wp-content/uploads/2018/09/AF-FCWG-Toolkit-Digital.pdf
[93] https://www.rggi.org/sites/default/files/Uploads/Proceeds/RGGI_Proceeds_Report_2016.pdf

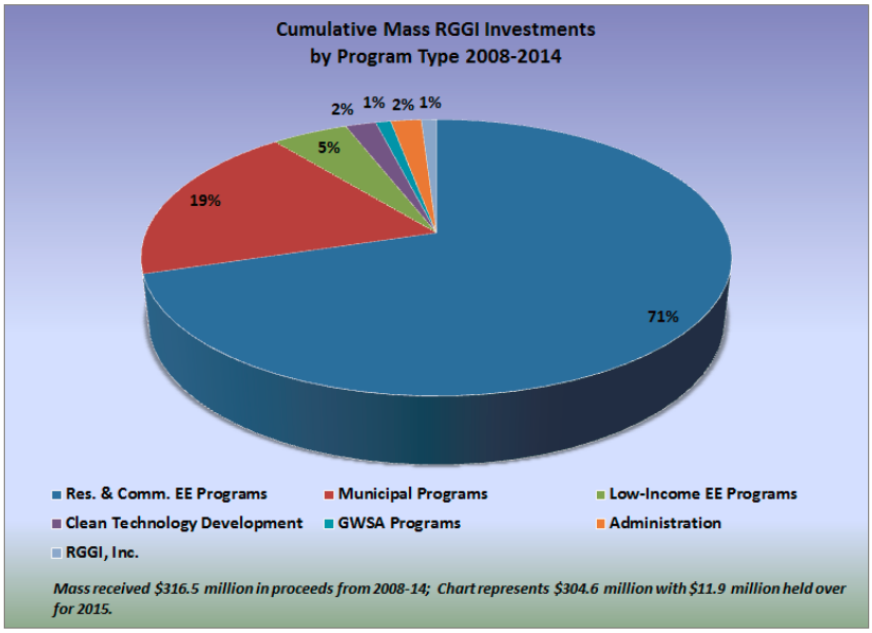
Benefits of 2016 RGGI Investments

Category	Annual Benefits of 2016 Investments	Lifetime Benefits of 2016 Investments
Participating Households	182,135 (Programs)* 806,605 (Bill Assistance)	N/A
Participating Businesses	2,680 (Programs)* 109,220 (Bill Assistance)	N/A
Workers Trained	370	N/A
Short Tons CO_2 Avoided	382,266	6.4 million
Equiv. Cars Off Road	74,258	1.2 million
Megawatt-Hours Saved	409,630	7.0 million
MMBtu Saved	1.6 million	30.4 million
Energy Bill Savings	$150.5 million	$1.7 billion

*Participants in all programs other than direct bill assistance.

RGGI in Massachusetts[94][95]

Since 2008, auctions have generated $470 million for Massachusetts. Massachusetts has reinvested $372 million in auction proceeds to increase the energy efficiency of residences and businesses, provide clean-energy solutions to 210 "Green Communities,"[96] and support the implementation of alternative energy resources.



RGGI auction proceeds have helped Massachusetts earn the nation's top ranking for energy efficiency six years in a row. By statute, the state must invest at least 80 percent of its allowance proceeds in energy efficiency. The remaining 20 percent of net funding has been used for Massachusetts' Green Communities to implement clean energy projects including energy efficiency improvements, or more recently for incentive programs for electric and plug-in hybrid vehicles.[97] Land-based natural resources absorb 14 percent of CO2 emissions and have the potential to capture at least 20 percent annually. No RGGI state, including Massachusetts, has used the afforestation offsets allowed under the program.

Between 2008 and 2018, total auction proceeds generated over $470 million. Twenty percent of this is $94 million, which could be used for nature-based climate solutions to sequester and store carbon and/or to help the Commonwealth adapt to a changing climate.

California Cap-and-Trade Invests in Natural Climate Solutions

Passed in 2006 and building on the passage of RGGI in 2005, Assembly Bill 32 requires California to return to 1990 levels of greenhouse gas emissions by 2020. All programs developed under AB 32 contribute to the reductions needed to achieve this goal and will deliver an overall 15 percent reduction in greenhouse gas emissions compared to the "business-as-usual" scenario in 2020 if we did nothing at all.

The Cap-and-Trade Program is linked with the program in Québec. It functions as a single market, although any proceeds from the sale of California allowances at auction are returned to California,

[94] Excerpted from https://www.rggi.org/
[95] Global Warming Solutions Act 5 year Progress Report for Massachusetts http://www.mass.gov/eea/docs/eea/gwsa/ma-gwsa-5yr-progress-report-1-6-14.pdf
[96] www.mass.gov/news/baker-polito-administration-designates-25-cities-and-towns-as-green-communities
[97] https://www.rggi.org/sites/default/files/Uploads/Proceeds/RGGI_Proceeds_Report_2016.pdf

and Québec proceeds are returned to Québec. California covers about 80 percent of statewide emissions – with the exception of emissions from natural and working lands. Entities with more than 25,000 metric tons CO2e annually are required to be in the program, as are all electricity importers. Essentially, that means the program covers all transportation fuel suppliers, electricity generators and electricity importers, and all large stationary, industrial sources. In total, the program covers more than 500 emitting facilities/sources, which are consolidated into approximately 350 entity accounts. Some companies will have multiple facilities under their control. [98]

The program offers more than half of emission permits (called allowances) for sale at quarterly auctions; entities are not required to participate in auctions, but the auctions have thus far resulted in approximately $11.8 billion to the State. With an increasing auction floor price, as well as the optional nature of the auctions, annual revenue cannot be predicted.[99]

The proceeds from auctions are appropriated through the budgeting process –pursuant to legislative and Governor priorities. Three-year investment plans are produced and guidance to assist in this process. California's program devotes millions to natural climate solutions. Cap-and-trade revenue is being invested into the following NCS-enhancing program areas: climate adaptation and resiliency, forest health, urban and community forestry and greening, wetlands and watershed restoration and wildfire prevention.[100] For example, the Department of Transportation receives five percent annually for low carbon transit operations. Other investments are made through the planning and budgeting process. Sustainable forestry has received $627 million to date, while climate adaptation and conservation easements through the San Francisco Bay Conservation Development Commission has received $20 million to date.

Georgia Forest Land Protection Act[101]

In Georgia, private land of at least 100 acres used to produce trees, timber, or wood products is eligible for a complete property tax exemption. The property may have secondary benefits such as wildlife habitat; carbon sequestration; mitigation and conservation banking of wetlands; and ecosystem services such as, clean air and water. The enrollment period is 10 years and is government by an agreement between the taxpayer and municipality.

Georgia also has a carbon sequestration registry, a non-profit program administered by the Georgia Forestry Commission and the Georgia Superior Clerks Cooperative Authority. The registry was established in 2004 and was based on the protocol of the Chicago Climate Exchange (CCX). The purpose of the Registry is to provide forest landowners, municipalities, and public and private entities with an official mechanism for the development, documentation, and reporting of carbon sequestration projects undertaken in Georgia. Participation in the Registry is completely voluntary. After initial development, the Georgia registry did not evolve to meet today's standards. The CCX closed around 2010, so the current methodology has been out-of-date for a long time.

[98] https://ww3.arb.ca.gov/cc/capandtrade/guidance/cap_trade_overview.pdf, personal communication with program director
[99] https://ww3.arb.ca.gov/cc/capandtrade/auction/proceeds_summary.pdf
[100] http://www.caclimateinvestments.ca.gov/
[101] https://dor.georgia.gov/georgia-forest-land-protection-act

There are only six projects currently in the registry, which have been listed since the mid-2000s. The projects include one private landowner, four federal land projects, and two state forests. They can still create a new project just for recognition on our registry website, but there is a fee. It would not pay for carbon credits, and the landowner would still have to go through a project developer elsewhere to establish credits. For the carbon registry certification, they held courses for those interested back in the mid-2000s. They have not had any certification course for the registry since at least, 2011.

Georgia only receives a few carbon sequestration requests each year, which are typically asking how to receive compensation for credits. Considering the complexity of today's carbon market and the fact the Georgia protocol is outdated, the registry is underutilized. If someone is interested, the state provides an overview of the carbon market and recommend landowners with 1,000+ acres contact a project development company for further assistance. The GFC will likely be too limited in funding, personnel, and interest to bring the registry up to speed going forward.

Tribes and Native-owned corporations are making huge profits by conserving their forests for carbon offset programs—an effort that could revolutionize conservation. In March 2018, Sealaska, a for-profit logging company collectively owned by nearly 23,000 Native Alaskans from the Haida, Tlingit, and Tsimshian tribes in Alaska's southeast islands, received approval to participate in California's cap-and-trade marketplace. The corporation will preserve 165,000 acres—45 percent of the forestland it controls—for 110 years to serve as a carbon sequestration bank. In return, Sealaska can sell carbon-offset credits to California companies that must curb their emissions under state law. This represented the first time public land was included in the carbon market. Sealaska's carbon-offsetting investment is expected to offset 11 million metric tons of carbon—the amount that 2.36 million cars emit in a year. With current offset prices at about $12 per ton, the credits could generate more than $100 million in revenue.[102]

Carbon Sequestration on School Trust Lands

Minnesota Office of School Trust Lands (OSTL) is in the process of discussing a public-private partnership to study carbon sequestration potential on Minnesota school trust lands using private grant dollars. The Minnesota DNR studied carbon market opportunities about five years ago. Their report outlines the existing carbon offset marketplace and identified potential opportunities for MN DNR to engage in this market. The report enacted a hypothetical forest carbon offset project on Minnesota DNR lands that was evaluated using the spreadsheet-based cost/benefit analysis tool. They evaluated two scales of projects:

1) 31,637 acres based on actual data from MN DNR lands
2) 100,000 hypothetical acres

The hypothetical MN project helps to demonstrate which factors are important in making a decision to develop an offset project, as well as shows the sensitivity of the revenue projections to key variables. Some major themes emerged:

[102] Excerpted from How to Make Millions While Saving a Forest, by Jake Bullinger May 7, 2018 Outside Magazine

- In all cases, a 20-year extended rotation and highest carbon price scenarios yielded the greatest net revenue over the life of the project.
- Climate Action Reserve (CAR)/Air Resources Board (ARB) scenarios were consistently lower than American Carbon Registry (ACR) and Verified Carbon Standard (VCS) scenarios because of the long-term nature (100 years) of the project which requires more capital investment in monitoring and reporting over the long term.
- Under all but the highest price scenarios, a CAR/ARB project on the 31,637-acre project fails to yield a net revenue. This is mitigated considerably on the 100,000-acre project where more offset credits are generated, and costs do not increase linearly.
- A third scenario with low inventory costs and higher carbon offset volumes yields the highest net revenue, particularly for VCS where the project period is shorter and long-term expenses are less (at both project acreages).
- At the $10 per ton offset price point, only VCS yields greater than $100,000 in net revenue for the 31,637-acre project and over $1,000,000 on the 100,000-acre project.

Carbon Incentives for Minnesota Landowners

In addition to property tax breaks, states can create or expand other financial incentives for landowners engaging in conservation, reforestation and improved management for carbon sequestration and storage purposes. Under this approach, policymakers could offer grants, loans, tax credits or deductions, as well as education, outreach and technical assistance to engage and support landowners.[103]

Minnesota could develop new incentives for landowners for improved management and stewardship strategies that optimize the carbon benefits of their farms and forests. For example, a Sustainable Forest Carbon Incentive program could provide qualified forest landowners with annual reimbursements from the state to reward carbon mitigation benefits. The program could be based on prescribed forest management practices on a per-acre basis, calibrated for different forest and tree types.[104]

A verifiable program could offer larger payments for highly productive forests with exceptional carbon forestry best practices. Landowners that engage in reforestation projects could receive a per acre tax incentive the first year, calibrated to the carbon sequestration rate for the type of forest, and then receive another payment per acre in subsequent years by demonstrating survival rates of the planted seedlings. In turn, the state could provide technical assistance that identifies which practices are appropriate for carbon purposes and could implement an incentive across a range of landowners committing to specific management practices.[105]

[103] https://www.state.sc.us/forest/mcs.htm
[104] Minnesota could utilize the Forest Inventory and Analysis (FIA) Program of the U.S. Forest Service provides annual survey data on forest tracts across the U.S., and how forests are likely to appear in the future, enabling the agency to evaluate whether current forest management practices are sustainable in the long run. https://www.fia.fs.fed.us
[105] Forest Climate Working Group: Tapping into U.S. Forests to Mitigate Climate Change http://forestclimateworkinggroup.org/wp-content/uploads/2018/09/AF-FCWG-Toolkit-Digital.pdf

Clean Water Act Funding

Under the federal Clean Water Act (CWA), the U.S. Environmental Protection Agency (EPA) awards capitalization grants to states to create and maintain State Revolving Funds (SRFs). EPA capitalization grants fund both the Clean Water State Revolving Fund (CWSRF) and the Drinking Water State Revolving Fund. States are required to match these capitalization grants with 20 percent of their own funds. Through section 604(b) of the CWA, states allocate CWSRF funding specifically to water quality management planning. CWA Section 319(h) funds are provided to designated state and tribal agencies to implement their nonpoint source management programs.

Clean Water State Revolving Fund (Section 212) [106]

The CWSRF program provides low-interest loans for water quality improvements. Through FY18, $133 billion has been provided to communities throughout the U.S. to protect public health, preserve valuable aquatic resources, and meet environmental standards.

Under the CWSRF, the EPA provides annual capitalization grants to states. In FY17, the annual capitalization grants totaled $1.37 billion. States are required to provide a 20 percent match. States use these capitalization grants to provide loans to public and private borrowers, including municipalities, inter-municipal, interstate, or state agencies, or qualified nonprofit entities. Interest rates must be at or below market rate, including interest-free, with a maximum term of 30 years. In some instances, additional subsidization can be provided in the form of principal forgiveness, negative interest rate loans, or grants.

Each year, states file an Intended Use Plan (IUP) with the EPA that describes how they will allocate their CWSRF funds. CWSRF funding has traditionally been used for wastewater treatment upgrades, though surface water protection and restoration and habitat protection and restoration are eligible activities. Building off of the Green Project Reserve, which was established in 2009, and requires funding for alternative, green projects, the EPA established the Green Infrastructure CWSRF Policy in 2016 to promote activities such as the protection of natural areas. Allocating CWSRF to natural climate solutions would be a good investment that both achieves the mission of the revolving fund in MN while mitigating carbon pollution.

The allotment formula for states has been up for nationwide debate for many years. To date it has included factors like population, growth rate, capital needs, and surveys. The program has provided money to states whether they need the funding or not, and thus an "ability to pay" set-aside has been discussed.

Since 1987, Minnesota has spent nearly $266 million on nonpoint source projects mostly around sewage treatment as well as best management practices for cropland and animals, which could provide natural climate solution benefits. In 2018, close to $4.5 million was spent on land conservation as well as habitat restoration. Minnesota received a total of $67.68 million in 2018.[107]

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[106] *Clean Water State Revolving Fund System*. A Trust for Public Land report by Matt Zieper, Brian Shillinglaw, Pat O'Connell and Honora Freeman.
[107] https://www.epa.gov/sites/production/files/2019-02/documents/mn.pdf

Both in Minnesota and nationally, only six to ten percent of SRF funding has been dedicated to nonpoint source projects.

The CWSRF system could be a potential source of funds for land conservation. CWSRF loans can be used to finance land management programs or land acquisition, including the purchase of a conservation easement, to protect the quality of streams, rivers and other local drinking water sources. Each state has great leeway to establish CWSRF programs that can support land conservation but lack of loan repayment sources for land conservation is a significant barrier. The land conservation and restoration projects that have been supported by state CWSRF programs to date have typically involved a public water utility with ratepayers, a local government with a pre-established dedicated source of conservation or restoration funding, or a sophisticated land conservation nonprofit like The Nature Conservancy or The Conservation Fund that can support the loan repayment through general organizational revenues or a specific revenue stream such as timber sale proceeds.

States could in many cases achieve NPS goals with NCS programs for the same amount of investment by targeting agricultural and forestry BMPs, including avoided conversion efforts. It could also support avoided conversion efforts. For example, Georgia and Ohio offer two varied and effective models for using the CWSRF for land conservation. Several other examples are outlined below.

Other approaches would require federal or state legislative action to modify water pollution permitting programs to either: a) give permittees the option of meeting a portion of any required pollution abatement through financing nonpoint source pollution abatement projects, which could in turn be financed through a CWSRF loan to that permittee; or b) require permittees borrowing from the CWSRF system to finance a point source pollution abatement project to borrow an additional 10 percent of the capital sought to finance nonpoint source pollution abatement. In either case, repayment would be financed by ratepayers of the permittee. This would eliminate the credit risk, transaction cost, and other key barriers to increasing CWSRF investment in NPS pollution.

Georgia Land Conservation Program
The Georgia Land Conservation Program was established in 2005 and capitalized with a one-time transfer of $55 million from the CWSRF. The Georgia Environmental Finance Authority administers this revolving loan program that provides low (0.9 percent currently) interest loans to local governments, nonprofit land conservation groups and state authorities. In total, the state has made $42 million in loans across 14 conservation projects, totaling 35,000 acres. Sample loan projects include:

- **City of Statham:** Loan of $135,000 in conjunction with a grant of $299,000 which protects the city's water source by adding 20.45 acres of mature hardwood forest around Statham Spring;
- **Decatur County:** Loan of $3,000,000 to help create the Silver Lake Wildlife Management Area, which provides habitat to endangered red cockaded woodpeckers and contains the unique longleaf pine ecosystem; and
- **Bibb County:** Loan of $180,000 in conjunction with a grant of $517,000 which protects an important bottomland hardwood in Tupelo Swamp.

Ohio Water Resource Restoration Sponsorship Program
The Ohio Water Resource Restoration Sponsorship Program (WRRSP) pairs traditional point-source pollution reduction projects with non-point source projects (conservation and restoration) and provides a significant interest rate subsidy to the point-source borrower who "sponsors" the NPS project.[108] [109] Since the WRRSP's inception in 1999, Ohio EPA has spent $160 million. In 2017, Ohio EPA will spend $10 million on WRRSP, or about one percent of the total $936 million CWSRF spending.[110]

Ohio is the only state that has seen regular utilization of its CWSRF to support land conservation projects – year in and year out. The Trust for Public Land's 2011 study of the Clean Water State Revolving Fund System recommends that states consider adoption of the Ohio sponsorship model as one proven means of deploying CWSRF capital to nonpoint source pollution abatement routinely and at scale. Through its Water Resource Restoration Sponsorship Program, Ohio has formed a linkage that brings together traditional CWSRF borrowers – public utilities – and pairs them with land conservation partners (both local governments and nonprofit) to carry out land conservation projects. Natural climate solutions like land conservation and restoration are bedrocks of this program. However, addition best management practices could be added. Some state SRF programs pass rules independent of the state legislature, while other take a more formal legislative route.[111] The Ohio program is successful because it builds upon the current strengths of the CWSRF system and provides financial incentives to traditional infrastructure borrowers to sponsor

[108] The loan recipient receives a reduced rate for their loan equal to the principal and interest costs of the project, plus an additional reduction of 0.1 percent as an incentive.
[109] The WRRSP is about protecting or restoring a resource itself; it is not part of a trading mechanism. Water or air quality trading is about offsetting adverse activity with mitigating activity elsewhere.
[110] The sponsor process is as follows:
- SRF advances loans equal to cost of Sponsor project from interest due in future on wastewater treatment plant (WWTP) loan to the borrower
- Borrower constructs WWTP improvements; Nonprofit / public Implementer conducts restoration and protection activities
- Borrower repays principal and remaining interest from WWTP to SRF
- No repayment by NPS Implementer to borrower or SRF for Sponsor project

[111] Personal communication with WRRSP program manager.

the land conservation projects. In order to increase state utilization of the CWSRF for land conservation, the Ohio program offers an excellent model that states can look to adapt.

Vermont Water Infrastructure Sponsorship Program (WISPr)
Following extensive consultation with Ohio regarding their WRRSP and how it operates, Vermont created WISPr in 2018 and is associated with their federally based SRF program. The program provides a funding source for natural resource projects (NRP) to encourage water restoration or protection projects that might otherwise not have been pursued. Sponsored nonprofit or municipal NRPs can include wetland restoration projects, floodplain/stream restoration, river corridor easements, woody buffer plantings, lake shoreline retrofit, erosion repair using LakeWise principles, dam removal. To be eligible, a municipal clean water project must sponsor an NRP and both projects will be funded with the same loan. The value of the natural resources sponsored project must be at a 1:10 or greater ratio to the municipal sponsoring project.[112]

The Town of Hinesburg, Vermont is involved in a $360,000 project called the Beecher Hill Brook Floodplain Restoration. This project will restore natural floodplain function and riparian habitat to a currently degraded and not hydraulically accessible area of the river corridor adjacent to a historically confined and incised stream section of Beecher Hill Brook. The Hinesburg Garage and a solid waste district operation have been recently relocated out of the river corridor and this project will restore the former operations area by removing fill and berms, planting riparian vegetation, and adding stream habitat features to the stream. This floodplain will improve water quality, habitat, and reduce flood risk, specifically by reducing velocity, reducing erosion, and capturing sediment, and cooling water by shading.

CWSRF in Minnesota[113]

The Clean Water Partnership (CWP) loan program is administered by the Pollution Control Agency (MPCA) to help local units of government address nonpoint source pollution problems. Loans can be made to counties, cities, watershed districts, and joint powers organizations that have an approved CWP diagnostic study/implementation plan or equivalent report. Loan funds may be used to implement a variety of best management practices (BMPs) such as sedimentation basins and detention ponds, replacement of failing individual sewage treatment systems, lakeshore landscaping for erosion control and stabilization, streambank stabilization, in-stream and inlake chemical treatment and aeration, forestry and parkland BMPs, and groundwater aquifer recharge area BMPs.

There is currently over $15 million in loan money available for which application can be made at any time. The interest rate on CWP loans recently was changed to 0 percent from 2 percent in an effort to increase participation. The loan term may vary depending on the project. These loans are used regularly for natural climate solutions programs and projects.

[112] https://dec.vermont.gov/facilities-engineering/water-financing/cwsrf/WISPr
[113] Minnesota Public Facilities Authority, Clean Water Revolving Fund 2019 Intended Use Plan, p. 12 and Minnesota Public Facilities Authority, Financial Report FY 2018, p. 5.

A 2013 project on the Sauk River call the Crooked Lake Basin Restoration project will use a CWP loan fund of $665,000 to purchase 1,280 acres from 12 landowners in order to install a ditch weir which will hold back water to reestablish the lake basin. Under management, a restored Crooked Lake will provide 2,200 acres of submerged and emergent plants that will filter and take up nutrients and stabilize sediments from Judicial Ditch 2.[114]

Loan repayments are in excess of $3 million per year, which are available to make new loans. MPCA staff have determined that funds are revolving at a level sufficient to meet current loan demand. Therefore, the MPCA has confirmed that it will not request a new allocation in 2019. The need for new money for the program will be reevaluated in 2020. Minnesota is among the 15 states that rank highest in NPS lending (either in percentage or total value terms) – Alaska, Arkansas, California, Delaware, Iowa, Maryland, Minnesota, Montana, Nebraska, New Hampshire, New Jersey, New York, Ohio, Washington, and Wyoming.[115]

The state Legislature did not appropriate funding for a CWP grant program in the 2016-17 biennium. There are no plans to open a CWP grant funding round in the immediate future. During the 2015 application cycle, $1,663,529 in grant money was awarded along with $2,748,800 in SRF loan funds. The legislature could continue to fund these grant opportunities for natural climate solutions.

The Minnesota CWSRF Green Project Reserve (GPR) guide defines green infrastructure to include, on a regional scale, the preservation and restoration of natural landscape features, such as forests, floodplains and wetlands. Fee simple purchase of land or easements on land that has a direct benefit to water quality, such as riparian and wetland protection or restoration is eligible for funding.[116] Until now funds have mostly used for green infrastructure, energy efficiency, some renewable energy projects. Funding cannot be used for avoided deforestation but could be used for reforestation if it had a water quality improvement component. One example would be adding tree trenches in parking lots to help with stormwater runoff and to provide shade for people and animals.[117] GPR eligible projects from 2018 and 2019 totaled close to $40 million.[118]

Other Source Water Protection

When pursuing public funding for natural climate solutions, a critical co-benefit is protecting drinking water supplies – and the most sustainable, affordable and effective way to protect drinking water sources from contamination is to conserve and restore watershed lands, especially aquifer recharge areas and wellheads and land that buffers streams, rivers, and lakes.[119]

The U.S. Environmental Protection Agency estimates Minnesota will need $7.4 billion in infrastructure repairs, upgrades and improvements in the next 20 years just to comply with the Safe Drinking Water Act. Shifting land use patterns, climate change and other factors present ongoing

[114] https://www.pca.state.mn.us/sites/default/files/wq-cwp8-22.pdf
[115] *Clean Water State Revolving Fund System*. A Trust for Public Land report by Matt Zieper, Brian Shillinglaw, Pat O'Connell and Honora Freeman.
[116] https://www.pca.state.mn.us/sites/default/files/cw-grn-guidance.pdf
[117] Personal communication with Minnesota Pollution Control Agency
[118] https://mn.gov/deed/assets/cw-iup_tcm1045-313601.pdf
[119] The Trust for Public Land's Source Water Protection Handbook, https://www.tpl.org/sites/default/files/cloud.tpl.org/pubs/water_source_protect_hbook.pdf

threats.[120] Natural climate solutions are not only cost-effective, but they also deliver a number of important in-state co-benefits. They improve water access and security, bolster biodiversity and support livelihoods.[121] Source water protection plans are a tool to keep drinking water clean and abundant. As of 2017, protection plans in Minnesota covered 91 percent of the population that gets water from a public water supplier, 513 communities, and 1.28 million acres of land.[122]

Statewide examples

Rhode Island "Penny Per Hundred"[123]
"Penny per hundred" describes Rhode Island's Public Drinking Water Supply System Protection Program.[124] The program collects nearly one penny (0.029 cents) for every one hundred gallons of water delivered from the state's water suppliers; 98 percent of Rhode Island residents are served by one of the state's 30 largest water suppliers.[125] The state then purchases bonds that are used for source water protection, as well as water quality improvement projects.[126]

For example, the Pawtucket Water Supply Board has received funding through the program to protect watershed lands that naturally filter pollutants and improve water quality by slowing surface runoff and increasing the infiltration of water into the soil. The result is less flooding and soil erosion, cleaner water downstream, which helps reduce treatment costs, and greater groundwater recharge.[127]

Not less than 55 percent of monies shall be spent for the acquisition of land or rights in land or physical improvements to acquired land required to protect the quality of raw water in the water supply system.[128] The Rhode Island Water Resources Board (WRB) administers the program, and supports the development, protection, conservation, and use of the state's water resources while providing for economic development and protection for the environment.[129] About 36 percent of surcharge collections had been remitted to the Rhode Island State Infrastructure Bank, which expended over $15 million to protect nearly 3,000 acres since 1994. According to the WRB, together with the Rhode Island Infrastructure Bank, the state will institute a new funding round in 2019 for funds that have been collected since 2015.

Vermont Clean Water Fund
The Vermont Clean Water Fund was created by Act 64, also known as the Vermont Clean Water Act, which was passed into law in 2015. The law also established a Clean Water Board to administer the Fund. The purpose of the Clean Water Fund is to provide additional state funds to

[120] Minnesota's drinking water: A treasure worth protecting Elk River Star News January 3, 2019
[121] Excerpted from https://www.nature.org/en-us/about-us/who-we-are/our-people/mark-tercek/innovative-solutions-for-nature-and-climate/
[122] http://www.health.state.mn.us/divs/eh/water/swp/about/index.html
[123]https://law.justia.com/codes/rhode-island/2016/title-46/chapter-46-15.3, http://www.wrb.ri.gov/work_programs_dwp.html
[124] Chapter 46-15.3-5 of Rhode Island general law, the "Water quality protection charge".
[125] Among these suppliers there are 14 municipal departments; two regional water authorities, one serves 3 towns and the other serves 4 towns and parts of 4 other municipalities; four quasi-municipal authorities with their own governing boards; and fifteen special districts created by the General Assembly. Rhode Island Office of Local Government Assistance, "Water governance in Rhode Island." 2007. http://www.municipalfinance.ri.gov/documents/resources/WATER%20%20GOVERNANCE..2007.pdf
[126] RI Gen L § 46-15.3-5 (2012)
[127] http://www.pwsb.org/content/land
[128] RI Water Resources Board, "Programs: Public Drinking Water Protection" http://www.wrb.ri.gov/work_programs_pdwp.html
[129] RI Gen L § 46-15.3-5 (2012)

help municipalities, farmers and others to implement projects including natural climate solutions, like adding forested buffers to cropland and pastures that will reduce pollution washing into Vermont's rivers, streams, lakes, ponds and wetlands. The fund is supported by a 0.2 percent surcharge on the property transfer tax on properties over $100,000. The Legislature added a new funding source, unclaimed bottle deposits, which will be effective January 1, 2020. In addition, the State Fiscal Year 2020 Governor's budget proposes revenues from the Estate Tax, which would be effective starting July 1, 2019 pending Legislative approval. The current estimated 2020 Clean Water budget is $27 million.[130]

Current Use

High business expenses may compel private landowners to convert their forests and farms to more profitable land uses or to harvest timber prematurely.[131] Tax liability can be the difference between choosing to permanently conserve land, a natural climate solution, or sell to a developer for immediate financial gain. Minnesota already provides several financial incentives for private landowners to conserve and manage forests and farms and could extend those incentives to encourage reforestation and improvement management practices that increase forest carbon sequestration and storage.

Current use taxation is the principal of taxing land at its current use value rather than its potential use value. Such programs provide property tax relief for owners of agricultural/rural/forested property in areas where land values are rising due to development or other factors. These programs could be very important if forest carbon values were another factor taken into consideration.

Minnesota offers four property tax relief programs that could be used to further incentivize carbon-reduction management or sequestration practices on agricultural and forestlands:

- Green Acres Program (agriculture)
- Rural Preserve Program (agriculture)
- Sustainable Forest Incentive Act (forests)
- Managed Forest Land (forests)

Green Acres Program[132]

The Green Acres Program provides property tax relief for owners of agricultural property in areas where the market value of land is affected by development pressure, sales of recreational land, or other nonagricultural factors.

Under the Green Acres program, taxes on enrolled properties are calculated on both the estimated market value (higher value based on highest and best use) and the agricultural value (lower value). The difference of the two calculated tax amounts is deferred until the property is sold or no longer qualifies for the Green Acres program. Only property that is classified as class 2a agricultural land is eligible for enrollment in the Green Acres program. The property must: be at least 10 acres in

[130] https://dec.vermont.gov/sites/dec/files/wsm/erp/docs/2010-02-04_CleanWaterFundFactsheet.pdf
[131] https://www.wri.org/sites/default/files/forest_management_indicators.pdf
[132] Minnesota Revenue, Green Acres Program, Property Tax Fact Sheet 5, Revised February 2018

size or a nursery or greenhouse; and be primarily devoted to the production for sale of agricultural products.

Currently the Green Acres provisions deferred $2,843,557,500 of approximately $14,739,566,700 in 2a-Agricultural EMV on 3,097,970 qualifying acres.[133]

Rural Preserve Program[134]

The Rural Preserve Program provides property tax relief for qualifying owners of rural vacant land that is part of the Green Acres program. Rural Preserve provides property tax relief to taxpayers with eligible rural vacant land that is part of a farm, by allowing the property to be valued at an agricultural value. Under the program, property taxes are based on that agricultural value and not value that is influenced by outside factors such as urban sprawl or seasonal activities.

Property that is classified as class 2b rural vacant land that is part of a farm is eligible for Rural Preserve. To qualify, the land must be part of an agricultural homestead, at least part of which is enrolled in Green Acres; not be enrolled in Open Space, metropolitan agricultural preserves, or the Sustainable Forest Incentive Act; and not have delinquent taxes.

The Rural Preserves program deferred $614,747,800 of approximately $1,302,775,400 in 2b-Rural Vacant EMV on 505,572 qualifying acres.

Sustainable Forest Incentive Act[135]

The Sustainable Forest Incentive Act (SFIA) provides payments to encourage sustainable use of forest lands. Property owners with qualifying lands are eligible to enroll in this program and can receive an incentive payment for each acre of qualifying forest land enrolled in SFIA. In return, property owners agree not to develop the land and to follow a forest management plan while in the program. All enrolled land must remain in SFIA for no less than the length of the covenant. This program has the full support of the timber industry in Minnesota who have thousands of acres enrolled in the program.

Qualifying properties must:

- Be current on property taxes
- Have 20 or more contiguous acres
- Be at least 50 percent forest land[136]
- Have a registered forest management plan in place
- Have an agreement ("covenant") in place limiting the property's use to forest management activities

Further, qualifying properties cannot:

- Be classified as 2c Managed Forest Land by the assessor or enrolled in:

[133] Personal communication with Minnesota Department of Revenue
[134] Minnesota Revenue, Rural Preserve Property Tax Program, Property Tax Fact Sheet 6, Revised March 2018
[135] Minnesota Revenue, SFIA, Property Tax Fact Sheet 9, Revised December 2017
[136] Minnesota Statutes 88.01, subd. 7

  - Reinvest in Minnesota (RIM),
  - Conservation Reserve Enhancement Program (CREP)
  - Conservation Reserve Program (CRP)
  - Green Acres
  - Agricultural Preserves
  - Rural Preserves

- Be used for residential or agricultural purposes
- Be improved with a structure, pavement, sewer, campsite, or road used for purposes that are not in the forest management plan
- Be covered under a Lessard-Sams Outdoor Heritage Council easement or similar easement granted after May 30, 2013.

Nothing in the law prohibits a landowner in SFIA to also receive a carbon credit payment, if that were an option. Any change to the law would require an act of the state legislature. Changes to this law have happened quite frequently during the budget process. Most recently the rate per acre was increased for most covenant lengths. With additional funding secured in recent years, total expenditures for SFIA would reach $10.9 million in FY18 and $11.2 million in FY19, an increase of over 90 percent in each year, and is expected to grow in subsequent years.[137] Changes to the law to include additional natural climate solutions would seem to be in keeping with recent changes as in the FY18-19 budget, changes were made which "modifies the existing SFIA program by adding to the act's purpose provision to emphasize economic and ecological benefits." The most recent change dealt with paved trails being implemented under enrolled forest land.

A forest management plan, often called a Woodland Stewardship Plan or Forest Stewardship Plan, guides the property owner's personal management actions to help achieve goals for the land. The plan is developed by a DNR-approved forester and is unique to the property. To receive SFIA payments, the property must be covered by a qualified forest management plan registered with the Minnesota Department of Natural Resources (DNR). The plan must have been written or updated within the last 10 years and prepared by a DNR approved plan writer. The property owner must follow the plan requirements while enrolled in SFIA to continue receiving payments. The covenant prohibits any use of the property that conflicts with the SFIA program, such as:

- Developing the property
- Using the property for residential or agricultural purposes
- Not following the forest management plan

The payment rates are adjusted each year based on average market values and tax rates but will not increase or decrease by more than 10 percent from the previous year. Note: payments are considered taxable income.
There is room to educate about longer rotation forests and carbon sequestration. It could be an opportunity to make landowners feel good about their woodland to explain to them in their

[137] https://www.house.leg.state.mn.us/Fiscal/Download/1013, https://mn.gov/mmb-stat/documents/budget/2018-19-biennial-budget-books/governors-recommendations-january-2017/natural-resources-department.pdf

Woodland Stewardship Plan how much carbon their wood is storing. TPL's Carbon Map could be one way for plan writers to estimate this.

The standard in MN is that all plans must be updated every 10 years.

Improving Current Use Laws

Several property tax relief programs in the U.S. have minimum enrollment periods. To improve these programs and incentivize the long-term retention and improved management of land-based carbon sinks, enrollment periods could be extended. For example, Wisconsin's Managed Forest Land program's minimum enrollment period ranges from 25 to 50 years.[138]

Over 80 percent of current use laws in the U.S. have a withdrawal penalty, key to ensuring a program's long-term success.[139] Most penalties are assessed through a roll-back tax (the taxes a landowner would have been required to pay in the absence of the program for a specified number of preceding years.)[140] [141] In addition to collecting roll-back taxes, some states charge an interest penalty in case of land conversion. For example, if land is withdrawn prematurely from Indiana's Classified Forest and Wildlands program, then the owner must pay the county up to 10 years in back taxes, plus a 10 percent per year interest penalty, and a withdrawal penalty of $100 per withdrawal and $50 per acre. If payment is not made, then a lien is placed against the property.[142]

A few states provide heftier tax credits for forests certified by the Sustainable Forestry Initiative or Forest Stewardship Council. States could also require the submission of plans that verify and reward increased carbon sequestration and storage through reforestation and improved land management activities. About half of all state current use programs require a forest management plan, including Minnesota.[143] DNR's Forest Management Plan contains many natural climate solutions and strategies to deal with forest health and productivity such as to:

- Develop and maintain a better balanced and complete age class distribution for plant communities managed primarily with even-aged silvicultural systems, while at the same time allowing some stands to transition to older growth stages; and
- Thin overcrowded stands to improve vigor and reduce competition.

Minnesota does not currently verify or reward increased carbon sequestration.

[138] http://docs.legis.wisconsin.gov/statutes/statutes/77.pdf
[139] http://harvardforest.fas.harvard.edu/sites/harvardforest.fas.harvard.edu/files/publications/pdfs/Ma_LandUsePolicy_2013.pdf
[140] http://www.calhouncounty.sc.gov/departments/PDFs/ROLLBACK percent20EXPLAIN.pdf
[141] https://www.lincolninst.edu/sites/default/files/pubfiles/2056_1378_WP12DU1.pdf
[142] https://www.in.gov/dnr/forestry/files/fo-ClassifiedForestBrochure.pdf
[143] https://www.srs.fs.usda.gov/pubs/ja/2011/ja_2011_butler_001.pdf

Current Use Examples from Other States

New Hampshire Land Use Change Tax
New Hampshire's Constitution allows privately owned, undeveloped land to be taxed at a lower rate to encourage the conservation of farms, forests, certified tree farms and wetlands. Approximately three million acres are enrolled in current use statewide.[144] Every year, the state Current Use Board assesses tax savings for different land use types.[145] Owners who meet the criteria for responsible land stewardship receive lower assessments provided their land is at least 10 acres,[146] and have management plans. When land is removed from its current use, it is subject to a Land Use Change Tax (LUCT) equal to 10 percent of the "best and highest" use of the land.[147] Municipalities recapture some of the funds it would have collected if the land had not been enrolled in current use. In 2015, 147 New Hampshire cities and towns collected $5,019,066 from the LUCT. Of that, $3,026,929 was deposited into local conservation funds.[148]

Iowa Forest and Fruit Tree Exemption
In Iowa, forest and fruit-tree reservations receive full exemption from property taxes for eight years as long as the reservation is at least two acres, consists of at least 200 trees per acre, and remains in a five-year program enrollment plan.[149] Owners are prohibited from removing more than 20 percent of all trees; and any trees that are removed or die must be replanted within a year.[150] A forest reserve cannot be used for economic gain other than raising trees, and livestock are prohibited. A working forest is an approved use. If the area is not maintained as a tree reservation during the eight years, or any of the five years following exemption, the landowner may have to pay the exempt taxes for up to five years.[151]

Lake Improvement Districts[152]

A Lake Improvement District (LID) in Minnesota is a political subdivision established by resolution of a county board, city council, or by the commissioner of the department of natural resources (DNR). A LID allows for greater local involvement in lake management activities. There are two ways in which a LID may be established:

1. By petition of property owners within the proposed district; or
2. By county board resolution.

Of the two, the petition process is the more common procedure. A petition requesting the formation of a LID must be signed by a majority of the property owners in the proposed district. The petition is addressed to the county board and filed with the county auditor. The county auditor verifies the signatures on the petition. A public hearing is held and the county board grants or denies the LID and if granted, sets the effective date for the LID to begin.

[144] https://forestsociety.org/advocacy-issue/current-use
[145] https://www.revenue.nh.gov/mun-prop/property/equalization-2016/documents/cu-alpha.pdf
[146] https://extension.unh.edu/resources/files/Resource000401_Rep423.pdf
[147] http://newenglandforestry.org/wp-content/uploads/2016/04/CurrentUseLaypersonsGuide.pdf
[148] https://www.revenue.nh.gov/mun-prop/property/equalization-2016/documents/cu-alpha.pdf
[149] http://www.co.marshall.ia.us/departments/assessor/forestfruit_html
[150] Ibid.
[151] Ibid.
[152] Minnesota Department of Natural Resources, Lake Improvement Districts - Questions and Answers

In lieu of a petition signed by a majority of the property owners, the county board may choose to initiate the establishment of a LID by itself. The board adopts a resolution declaring its intent to establish a LID. A public hearing is held, and the decision is made.

In its resolution establishing a LID, the county board delegates specific authorities to the LID. Each LID is granted different levels of authority depending upon existing problems and proposed activities. Some of these may include the right to:

1. Apply for and comply with DNR and other agency and local permits and policies required for specific lake management projects;
2. Undertake lake research projects;
3. Conduct programs of water improvement and conservation;
4. Serve as local sponsor/recipient for state and federal projects or grants;
5. Maintain public beaches, docks, and other public facilities;
6. Regulate water surface use;
7. Provide and finance certain local governmental services within the LID;
8. Implement a water-monitoring program.

Projects and activities undertaken by a LID must comply with all local, state and federal regulations. Additionally, a lake improvement district may not undertake projects on any lake unless public access is available.

A LID does not have any taxing authority of its own. Rather, the county board specifies the funding arrangement when it establishes the LID. Typically, a LID must pass its budget recommendations through the county board for approval and actual collection. To finance LID projects, services and general administration, a county may:

1. Assess costs to benefited properties;
2. Impose service charges;
3. Issue general obligation bonds;
4. Levy an ad valorem tax solely on property within the district; or
5. Any combination of the above.

A LID can also be the recipient of grants from state and federal agencies and private foundations.

LIDs provide an opportunity for education and for promoting activities by private landowners. Some examples include:[153]

- The Circle Lake Improvement District in Rice County was established in January 2019—this lake is in the Cannon River watershed and the founders have made promoting land restoration and conservation activities in the watershed as well as nutrient management and erosion control part of their mission.

[153] Personal communication with Kathy Metzker, Hydrologist, MN Department of Natural Resources

- The Lake Zumbro Improvement Association was formed to address sedimentation in Lake Zumbro. In doing so, it works closely with other organizations through the Zumbro Watershed Partnership, which is 'a non-profit organization working to improve land management and water quality in the Zumbro watershed.' This is an opportunity to promote land conservation activities that help the climate while also reducing sedimentation.

The Chisago Chain of Lakes LID is the oldest, largest, and most active LID in Minnesota. This LID includes the entire Chisago Chain watershed (making it unique) and has the ability to directly engage in landscape restoration and conservation projects. It was also formed by county resolution, and the county is active in oversight and management.

While lake improvement districts do not have taxing authority, they could use grant funds to implement natural climate solutions. The state could also provide incentives for LIDs that create climate management plans to includes specific NCS strategies and could target existing or new state funding as part of a matching grant program to support natural climate solutions by such districts.

Watershed Districts

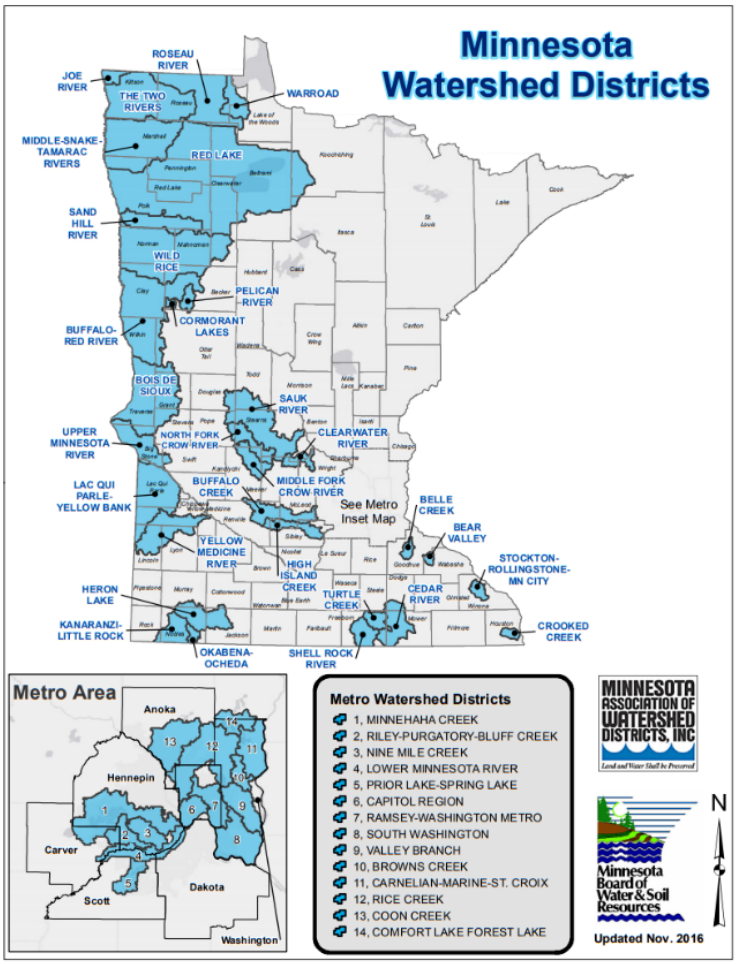
Watershed districts are local units of government that work to solve and prevent water-related problems. The boundaries of the districts follow those of a natural watershed (an area in which all water drains to one point).

Minnesota has 46 watershed districts, most of which are named after the primary lake or river within the watershed. Minnesota's watershed districts do not cover the entire state. They are created through a local petition process. The specific duties of Watershed Districts vary across the state -- some focus mainly on flood damage reduction, while others have a broad range of programs and services to protect and improve water quality. Each watershed district is governed by a board of managers appointed by the county boards of commissioners with land in the watershed district. Each watershed district is also required to have a citizen advisory committee to provide input to the managers on projects and activities.[154]

Minnesota created watershed districts in 1955, through the Watershed Act, with the idea that water management policies should be developed on a watershed basis. The purposes of watershed districts are to conserve the natural resources by land use planning, flood control, and other conservation projects by using sound scientific principles for the protection of public health and

[154] Minnesota Board of Water and Soil Resources

welfare and the provident use of natural resources. Under Minnesota law, watershed districts are authorized to:



- Adopt rules to regulate, conserve, and control the use of water resources within the district.
- Contract with units of government and private and public corporations to carry out water resource management projects.
- Hire staff and contract with consultants.
- Assess properties for benefits received and levy taxes to finance district administration.
- Accept grant funds, both public and private, and encumber debt.
- Acquire property needed for projects.
- Acquire, construct, and operate drainage systems, dams, dikes, reservoirs, and water supply systems.
- Enter lands within and without the district to make surveys and conduct investigations.

Watershed districts may levy:

- Up to 0.01596 percent of estimated market value, or $60,000, whichever is less, for organizational expenses and preparation of a watershed management plan for projects.
- Up to 0.048 percent of estimated market value, or $250,000, whichever is less, for general administrative expenses and for the construction or implementation and maintenance of projects of common benefit to a watershed district.[155]

[155] In addition to the annual general levy, the managers may annually levy a tax not to exceed 0.00798 percent of estimated market value for a period not to exceed 15 consecutive years to pay the cost attributable to the basic water management features of projects initiated by petition of a political subdivision within the watershed district or by petition of at least 50 resident owners whose property is within the watershed district.

- Up to 0.02418 percent of estimated market value for a survey and data acquisition fund.[156] The balance of the survey and data acquisition fund may not exceed $50,000.
- In addition, watershed districts may levy a tax: (1) to pay the costs of projects undertaken by the watershed district which are to be funded, in whole or in part, with the proceeds of grants or construction or implementation loans; (2) to pay the principal of, or premium or administrative surcharge, if any, and interest on, the bonds and notes issued by the watershed district; and (3) to repay the construction or implementation loans for projects under the Clean Water Partnership (CWP) Program.

Watershed districts are authorized to raise private and local funds and could use funds to implement natural climate solutions. The state could provide incentives for watershed districts that create climate management plans to includes specific NCS strategies and could target existing or new state funding as part of a matching grant program to support natural climate solutions by such districts.

A watershed district located wholly within the Metropolitan area[157] has the duties and powers of a joint powers watershed management organization, which includes the authority to adopt a budget and decide on the total amount necessary to be raised from ad valorem tax levies to meet the budget.[158] Thus it appears that watershed districts in the metropolitan area do not have a limit to the amount of ad valorem taxes they can levy.

For example, in 2018, the Bear Valley Watershed District levied $10,000. In contrast, the Capitol Region Watershed District (which is in the metropolitan area) levied $6.5 million.[159] Both entities have funded natural climate solutions, including the restoration of Lake McCarrons Shoreline and participating in a target watershed program focusing on water quality in Como Lake and Lake McCarrons (Capitol Region). There is also participation in extensive tree planting and participation in the state's buffer protection establishing new perennial vegetation buffers of up to 50 feet along rivers, streams and ditches that will help filter out phosphorus, nitrogen and sediment.[160]

Conservation Tax Credits and Incentives

State laws can provide tax incentives to private landowners who donate land or easements to public or private, nonprofits for conservation purposes. Tax incentive programs supplement other open space funding programs by encouraging private, voluntary land conservation. When combined with existing federal and state charitable deductions, tax credits may make conservation a more attractive option for landowners over development. Tax credits may be targeted to state specific objectives, such as forest or watershed protection.

[156] A survey and data acquisition fund is established and used only if other funds are not available to the watershed district to pay for making necessary surveys and acquiring data. The property tax can be levied only once every five years.
[157] The area over which the Metropolitan Council has jurisdiction, including only the counties of Anoka; Carver; Dakota excluding the cities of Northfield and Cannon Falls; Hennepin excluding the cities of Hanover and Rockford; Ramsey; Scott excluding the city of New Prague; and Washington. (Minnesota Statutes 473.121)
[158] Minnesota Statutes 103D.335 Subd. 23 and 103B.211
[159] Minnesota Revenue, Preliminary 2019 Property Tax Levies
[160] https://www.goodhueswcd.org/buffers

Most state tax incentives are structured as tax credits against a landowner's state income tax liability. For those landowners with little taxable income, some states allow the landowner to transfer or sell tax credits to another taxpayer and/or to carry the credit forward over a number of years. Several state tax incentives apply to fee simple donations as well as conservation easements. When state incentives are used in conjunction with the federal incentive for donations of conservation easements, voluntary land donations can become more affordable.

The following states currently have a statewide land conservation income tax credit program: Arkansas, California, Colorado, Connecticut, Delaware, Georgia, Iowa, Maryland, Massachusetts, Mississippi, New Mexico, New York, South Carolina, and Virginia.

The state does have the Agricultural Land Preservation Program statute (Minn. Stat. Ch. 40A) which makes it possible for owners of qualifying land to receive property tax credits of $1.50 per acre per year, and other benefits, in return for agreeing to preserve their farms for long-term agricultural use. Property tax credits are funded by a surcharge on mortgage and deed recordings, which are discussed later in this document.

Minnesota could use one or more of existing state incentive programs as a model(s), and create a new or enhanced state tax incentive for private landowners who are interested in NCS (via avoided conversion, etc.)

Colorado[161]
A conservation tax credit is available to Colorado residents, corporations, estates, and trusts that donate a conservation easement. The credit is valued at 75 percent of the fair market value of the first $100,000 of the donated value, and then 50 percent of any remaining amounts between $100,000 and $1.5 million per donation. In Colorado, a taxpayer can also sell all or part of a credit to a transferee, usually for 80-85 percent in cash. Credits may be carried forward for up to 20 years. In a year of a state budget surplus, a donor can also get up to $50,000 in cash refund for the credit. Since 2000, over 2.2 million acres has been conserved via easement using the tax credit.

Georgia[162]
The Georgia Land Conservation Program provides a state tax credit to individuals and corporations donating land or easements for conservation. The tax credit allows taxpayers to claim a credit against their state income tax liability of 25 percent of the fair market value of the donated property interest, up to $250,000 for individuals and $500,000 for corporations. Unused credits may be carried forward over 10 years. Donors can sell tax credits to other Georgia taxpayers. Lands enrolled in the Conservation Use Valuation Assessment program, also known as the 10-year program, are eligible unless they are otherwise restricted by some form of permanent protection. The aggregate amount of tax credits that can be awarded per calendar year is $30 million. The tax credit sunsets on December 31, 2021.

Virginia[163]

[161] https://www.colorado.gov/pacific/dora/conservation-easement-tax-credit-certificates
[162] https://glcp.georgia.gov/about-tax-credit-program-0
[163] http://www.tax.virginia.gov/content/whats-new-tax-credits

In Virginia, every landowner who donates land or an easement for conservation is entitled to a credit against state income tax worth 40 percent of the easement's fair market value. If the credit is not used up in the year of the easement donation, it can be carried forward for 10 to 13 years, depending on the year of donation. Furthermore, the credit may be sold or transferred to other Virginia taxpayers. There is a cap on the amount of credits an individual can claim in a single year, but since the credits are transferable a donor should still be able to claim full credit--they may just have to transfer credits to more separate taxpayers. For 2017, the individual cap is $20,000, and the cap is scheduled to go up to its previous amount of $50,000 for 2018.

Maryland Forest Conservation Act

Maryland's Forest Conservation Act[164] makes the identification and protection of forests and other sensitive areas an integral part of the development planning process.[165] Under this regulatory program, developers must gain approval at the county or municipal level for develop proposals, and are required to mitigate 75 percent of the tree cover that will be lost through reforestation, and/or pay a fee. The program has slowed net forest loss in the state, with at least twice as many acres protected and planted than have been cleared; however, the loss of mature trees and other issues has led advocates to recently try to make improvements to the law.[166]

North Carolina Forest Development Program

North Carolina's Forest Development Program (FDP) is a reforestation program under which a landowner is partially reimbursed for the costs of site preparation, seedling purchases, tree planting, and other work needed to establish a new forest. To qualify for reimbursement, forest owners must demonstrate they have an approved land management plan.[167] FDP reimburses up to 40 percent of the cost per acre, or 40 percent of the rate for management practices in the region, whichever is less. The program sets a 100-acre reimbursement limit per landownership group, meaning that a woodland owner that reforests 200 acres of land would only receive cost-sharing assistance on half of those acres.

Arkansas Southern Pine Beetle Prevention Program

The southern pine beetle has destroyed millions of acres of private land and caused billions in damages.[168] Arkansas offers an incentive to support improved forest management by encouraging landowners to engage in thinning when it otherwise would not be financially feasible.[169] Thinning reduces the risk of carbon-emitting wildfires and the threat of insect infestations. Woodland owners can receive payments of $50 per acre, up to $10,000 per fiscal year. To qualify, land must be non-industrial, private, at least 10 acres, and inspected prior to beginning the thinning program. A forester re-inspects sites after each logging operation to ensure it was adequately administered.[170]

New York

[164] https://dnr.maryland.gov/forests/Pages/programapps/newFCA.aspx
[165] http://marylandreporter.com/2018/01/05/saving-md-forests-again-on-state-house-environmental-agenda/
[166] http://www.mdlcv.org/forest-conservation-act
[167] https://content.ces.ncsu.edu/financial-incentives-for-forest-management
[168] https://www.fs.fed.us/foresthealth/publications/spb_success_story.pdf
[169] http://www.aad.arkansas.gov/forest-management-programs
[170] http://www.aad.arkansas.gov/forest-management-programs

Provides a state income tax credit equal to 25 percent of the property tax on a property protected by a conservation easement, in perpetuity. Capped at no more than $5,000 per year. This credit is refundable.

Mississippi
Provides a state income tax credit for 50 percent of a donor's transaction costs (survey, appraisal, title research) involved in a donation of a certain lands or conservation easements, up to $10,000.

Arkansas
The Wetland and Riparian Zones Tax Credit Program provides a tax credit of 50% of the value of restoration costs incurred to protect certain wetlands, capped at $50,000.

Connecticut
Provides a tax credit against the state business tax of 50 percent of the value of open space land donated by a corporate entity. May be carried forward up to 25 years.

Florida
Provides an exclusion of a conservation easement protected property from 100 percent of property tax.

New Incentives to Promote Natural Climate Solutions

Increasing Wood Product Usage
New technology utilizing "mass timber" panels, cross laminated timber (CLT), and other wood-based building systems has allowed wood to emerge as a cost-effective and sustainable material for construction.[171] Minnesota could increase the adoption of these new technologies and could explore a range of policy approaches. These options include public awareness programs, new procurement policies for government buildings, providing tax incentives for low-carbon building construction, and updating building codes.

Oregon is one of the leaders in this field at both the state and local levels. Officials with the state of Oregon have shown an interest in ramping up manufacturing of advanced wood products to support the state's rural timber towns. The governor's office, through Business Oregon and other state agencies, actively supports the development of advanced wood products markets and manufacturing. The Oregon Building Codes Division has been instrumental in getting permits issued for mass timber buildings constructed with advanced wood products. Some specific actions have been:

- A 2013 Executive Order entitled "Promoting Wood Products in Commerce as a Green Building Material, Encouraging Innovative Uses of Wood Products and Increasing Markets for Oregon Wood Products";[172]

[171] Pingoud K., Pohjola J., Valsta L, 2010. *Assessing the integrated climatic impacts of forestry and wood products.* Silva Fennica 44(1): 155–175
[172] https://www.oregon.gov/gov/Documents/executive_orders/eo_12-16.pdf

- In 2018, Oregon became the first state to legalize mass timber high-rises, and it could lead to a boom in tall wood buildings.[173]

Clackamas County outside of Portland, which encompasses large portions of Mt. Hood National Forest, wants to lead the growing CLT industry in Oregon by developing a supply-increasing pilot program that can spur rural economic development.[174] They are investing in the technology by:

- **Increasing supply:** by raising the amount of timber harvested in working forests on county public lands.
- **Providing technical expertise:** their Business Response Team can support advanced timber technology clients with staff expertise in mass timber, fire/life safety, structural engineering and more.
- **Encouraging investment:** by attracting mass timber investments with a market demand study, financial modeling, and local, state and federal incentives for investment in distressed communities. Specifically, they are pushing for New Market Tax Credits – federal incentives that prompt investment into distressed communities and keep jobs local.
- **Legislative advocacy:** by advocating for more forestland to be used specifically to provide supply for CLT.

Public awareness programs

States sometimes provide grants to organizations that promote forest products. One such example is WoodWorks, a national nonprofit that provides free technical support, education and resources related to the design of wood buildings. The organization has effectively educated builders, homeowners, architects and engineers about the benefits of forest products, while giving them the tools to reduce the costs of designing and constructing wood buildings.[175]

The group has worked on over a dozen projects in Minnesota. One example Schaar's Bluff Gathering Center in Dakota County. Located on the Mississippi River, this is a 3,500-square-foot environmental education center that emphasizes the stewardship of ecological and cultural resources. Referred to as a "living building," the project ranks within the top 1 percent of all sustainable structures. Among its green attributes, an on-site wind turbine generates most of the building's electricity and the high-performance envelope is super insulated. The building is constructed with ironwood and features pine, oak and ash in the interior. All of the wood used in the interior was locally harvested from sustainably managed forests.[176]

Current investments of $1 million in WoodWorks have reduced overall emission levels by 3.6 million tons of carbon per year, through a combination of increased carbon sequestration and avoided emissions.[177] State and local governments in Minnesota could expect similar impacts with investment in other public awareness and education programs.

[173] https://www.businessinsider.com/tall-timber-projects-around-the-world-2018-8
[174] https://www.clackamas.us/forests/clt
[175] http://www.woodworks.org/wp-content/uploads/About-WoodWorks-Wood-Products-Council-Oct-2017.pdf
[176] Excerpted from https://www.woodworks.org/project/2009-green-building-award/
[177] Forest Climate Working Group. 2015a. Expanding the Use of Wood in Buildings – Including Tall Wood Buildings – Helps Support Climate Preparedness and Mitigation. Supplement to Forest Climate Working Group Recommendations. January 14, 2015. 14 pp.

New procurement policies

Minnesota could establish procurement policies for government-owned and -funded buildings, requiring low-carbon materials in construction. Implementation of such a policy could reduce carbon emissions from the building-construction sector by about 9.5 percent.[178]

Minnesota's B3 – "Buildings, Benchmarks, and Beyond" – is a set of tools and programs for sustainable buildings in the state that became effective as of January 1, 2004. The B3 tools and programs are designed to help make buildings more energy efficient and sustainable. The B3 programs have been developed for and are required on State-funded projects in Minnesota, however, they are easily applied to any project. [179] The B3 Guidelines focus on meeting sustainability goals for site, water, energy, indoor environment, materials, and waste, and can be applied to new buildings or renovations. All projects that receive general obligation bond funding from the State of Minnesota are required to follow the B3 Guidelines. The Guidelines are also available to be used by any project voluntarily.

The 2019 Capital bonding bill contained millions of dollars across multiple agencies for building improvements. In addition to applying B3 standards, they could also require a portion of funding to be dependent on use of sustainable materials like wood.

Any project following the B3 Guidelines will also be automatically applying the Sustainable Building (SB) 2030 Energy Standard, which is focused on on-site renewable energy generation and energy efficiency. Minnesota's SB Energy Standard is a progressive energy conservation program designed to significantly reduce the energy and carbon in Minnesota commercial, institutional and industrial buildings. SB 2030 sets a new reduced total carbon emissions target every 5 years, with the aim of reaching net zero carbon by 2030.[180] Like the B3 Guidelines, the SB 2030 Energy Standard is required on all Minnesota projects that receive general obligation bond funding from the state that have started Schematic Design after August 2009, and can be used on a voluntary basis on any project.

Some Minnesota municipalities, such as Saint Paul, have their own sustainable building policies. The Saint Paul Sustainable Building Policy, adopted in 2010, requires that certain construction projects—new projects receiving more than $200,000 in public assistance, and certain rehab projects—be built to a high green standard, and provides complementary resources. The Policy was developed as a result of a 2007 grant from the Minnesota Pollution Control Agency for the city to develop a green building policy that could serve as a model.[181]

The 2019 Capital bonding bill contained millions of dollars across multiple agencies for building improvements. In addition to applying B3 standards, they could also require a portion of funding to be dependent on use of sustainable materials like wood.

[178] Forest Climate Working Group. 2015a. Expanding the Use of Wood in Buildings – Including Tall Wood Buildings – Helps Support Climate Preparedness and Mitigation. Supplement to Forest Climate Working Group Recommendations. January 14, 2015. 14 pp.
[179] https://www.b3mn.org/wp-content/uploads/2018/11/B3GuidelinesVersion31_20181112.pdf
[180] https://www.b3mn.org/wp-content/uploads/2017/07/SustainableBuilding2030factsheet.pdf
[181] https://www.stpaul.gov/DocumentCenter/View7/Sustainable%20Brochure%20for%20Web_201407241137536729.pdf

Tax incentives for low-carbon building construction
Minnesota could establish a tax credit for commercial and residential building owners that use low-carbon materials in their construction projects, as currently none exist in the state.[182] This credit could be awarded upon proof of installation, structured similarly to credits provided for energy-saving installations like solar panels and geothermal heating systems. The state could develop a list of low-carbon building materials that qualify for this program and offer tax credits commensurate with the carbon benefits of each product.

On the federal, level Section 45L of the Internal Revenue Code (IRC) offers up to $2,000 per unit credit for single and multi-family dwellings of three stories or less. Therein, an approved 100-unit apartment building could yield a $200,000 tax liability savings.

The qualifying prerequisite mandates that the structure must deliver heating and cooling consumptions levels lower than national energy standards. Because the credit can be claimed retroactively (from 2005) and many developers have been building to meet energy standards for the past few years, it's worth a 45L credit assessment to check for potential claims.

On the state level New Mexico offers a Sustainable Building Tax Credit (SBTC). The SBTC is an income tax credit to encourage private sector design and construction of energy efficient, sustainable buildings for commercial and residential use. The tax credit is based on third-party validation of the building's level of sustainability. The amount per square foot a homebuilder or homebuyer receives per home is a maximum of $6.50 and it caps the eligible size per home at 2000 square feet. Water conserving features are now required: each home must have indoor plumbing fixtures and water-using appliances that have average flow rates equal to or lower than the Environmental Protection Agency's "WaterSense" certification, and it requires a water line that can be connected to a drip irrigation system in the front and back of a residence in any potential landscaping areas.[183]

As of June 2019, New Mexico had over $2 million available for residential housing tax credits in 2020.

Updating building codes
Building codes – legal documents regulating the construction, maintenance and occupancy of buildings – can play a critical role in enhancing wood product usage. The International Building Code (IBC) has been adopted as the standard by most jurisdictions in the United State Amended versions of the IBC increasingly include more opportunities for wood construction, identifying fire protection techniques, and allowing wood usage in a wider range of building types.

However, the IBC still lags behind modern wood construction technology. For example, it does not explicitly allow for the usage of CLT in the construction of buildings taller than six stories. Therefore, the approval of mass timber buildings occurs on a case-by-case basis,[184] which often discourages timber projects. The IBC also sets a height limit for wood buildings, deeming all mass

[182] Personal communication the Minnesota Department of Employment and Economic Development
[183] http://www.emnrd.state.nm.us/ECMD/CleanEnergyTaxIncentives/SBTC.html
[184] https://urbanland.uli.org/sustainability/building-case-larger-wood-buildings-north-america/

timber buildings that exceed this limit "tall mass timber." Because CLT has demonstrated to be a fire resilient and sustainable alternative to other building materials, the International Code Council has appointed a committee of building officials, fire officials, and architects to examine and amend current code requirements.[185] Minnesota would need to ensure they approve these IBC amendments.

When used as a building material, wood has the unique ability to store large amounts of carbon for long periods of time.[186] The production of wood as a building material produces significantly lower levels of GHG emissions and uses far less fossil fuel energy than the production of other building materials. Both forests and wood are natural carbon storage systems. Creating 100 ft^3 of wood absorbs 2.4 tons of CO_2. 9.3 tons of carbon, or 34 tons of CO_2, are stored in the building structure of an average US single-family home. 1.5 billion metric tons of carbon equivalent to 5.4 billion metric tons of CO_2 are currently stored by US wood structures.[187]

Wood is the only renewable building material, it can be reclaimed and reused numerous times, and at the end of its lifecycle, the solar energy it stores can be recovered for energy production. In Minneapolis, T3, a 7-story, 224,000 square foot office building constructed of wood, will take the equivalent of 996 cars off the road for a year.[188] By means of the substitution effect, 483 million tons of CO_2 equivalent are avoided every year through the use of wood building materials. Additionally, the market for wood-based products provides an incentive for landowners to maintain land as forest, as opposed to other uses, thereby preventing the loss of large carbon stores.

The wood products sector is a top 10 manufacturer in 47 states. The US wood products sector employs about 900 thousand people, has a $50 billion payroll, and generates less than 1 percent waste in manufacturing. [189] The timber industry is Minnesota's 5th largest manufacturing industry employing over 40,000 workers and generating over $8 billion in forest products.

If Minnesota wanted to build upon its use of use of wood products the state or local governments could pass a building code reform law or a new loan, grant or incentive program to encourage developers to use wood products.

Case Example: El Dorado High School

In 2011, El Dorado High School in Arkansas became one of the first schools in the state to utilize wood in school construction. [190] The school had to expand to accommodate several hundred additional students and intended on completing the project using steel and masonry. However, the project's budget estimates surpassed $60 million, which the district could not afford. Motivated by a need to cut costs, officials turned to wood as an alternative construction material and spent just $43.2 million.

[185] http://www.awc.org/tallmasstimber
[186] http://www.dovetailinc.org/report_pdfs/2015/building_with_wood.pdf
[187] Ibid.
[188] https://www.thinkwood.com/wp-content/uploads/2018/03/Think-Wood-Infographic-Wood-and-Environmental-Benefits.pdf
[189] http://www.dovetailinc.org/report_pdfs/2015/building_with_wood.pdf, p.3
[190] http://www.woodworks.org/wp-content/uploads/2012/02/cs-el-dorado.pdf

The potential carbon benefit of the project was enormous. 4,340 cubic meters of wood were used in the construction of the building, an amount that can be grown in U.S. and Canadian forests in 13 minutes. The wood, used in place of fossil-fuel intensive materials like steel and concrete, prevented the emissions of 7,780 tons of carbon and sequestered 3,660 tons, for a total potential carbon benefit of 11,400 tons of carbon. This is the emissions equivalence of removing 2,100 cars off the road for a year or powering a home for nearly a thousand years.

El Dorado High demonstrated how using wood reduces construction costs while benefitting the environment. Future state-funded construction projects should emulate it in order to achieve similar success.

LOCAL FINANCING AUTHORITY

Federal and state conservation programs are limited, in high demand, and by themselves cannot meet land conservation needs. Accordingly, state-enabling legislation that allows communities to fill in the gaps for land conservation funding is crucial. When states provide local governments with the legal authority to tax and dedicate revenues for land conservation, local dollars and local control are expanded. Common local financing options include the property tax, local option sales tax, general obligation bonds, real estate transfer taxes, special assessment district fees, and budget appropriations.

Common Land Conservation Revenue Streams: Local Programs	
Revenue Stream	**Selected examples**
Property tax	Washington, New Jersey, Massachusetts
Sales tax	Arizona, Colorado, Florida, New Mexico
General Obligation bonds	Arizona, California, Colorado, Montana, New Mexico, Utah
Income tax	Pennsylvania
Real estate excise tax/fees	Washington, Maryland, New York
Source: The Trust for Public Land's Conservation Almanac	

Generally, states provide local governments with the legal authority to tax and dedicate revenues for land conservation and natural resource protection. The most common local funding option for open space conservation in Minnesota is general obligation bonds. Other options may be available such as sales taxes, which require approval by the state legislature.

Expanding State Incentives for Local Funding

Local governments are important partners in financing and preserving open space. With a reliable source of funds, cities, towns, and counties can establish meaningful conservation priorities that protect the most valuable resources and meet local and statewide goals. Local governments with significant funds are better positioned to secure and leverage funding from federal government and attract other state and private philanthropic partners.

States that have enabled expanded local financing options and provided state incentives, such as Massachusetts and New Jersey, have seen significant increases in the number of local jurisdictions that have undertaken or improved open space planning and established their own local funding programs, making them important partners in reaching statewide goals. In Minnesota, incentives could be specifically directed to communities in the Mississippi Headwaters.

For example, passed in 1989, New Jersey enables counties and municipalities to raise additional local funds by establishing voter-approved Open Space Trust Funds supported by local property taxes. All 21 counties and over 250 municipalities in New Jersey have established an Open Space Trust Fund via referendum. The tax may be levied at any amount.

Colorado's Open Space Sales Tax Exemption, passed in 2007, gives counties the authority to seek voter approval of a sales and use tax to fund open space programs. These open space programs fund projects that conserve and maintain natural areas, water quality, wildlife habitat, open space, and working farms and ranches. This program is a resource for financially constrained county governments who recognize the need to create conservation programs. There is no cap on the tax.

The Massachusetts Community Preservation Act, passed in 2000, combines local enabling authority with a commitment of state funds to urge communities to implement a local property tax for open space, historic preservation, and affordable housing. Cities and towns in Massachusetts are authorized to impose a surcharge of up to three percent on local property taxes. State matching funds are provided from a document-recording fee at the registries of deeds. Each year, the statewide CPA Trust Fund monies are divided among communities that have adopted CPA based on the total amount raised by the local CPA surcharge. To date, 175 municipalities have adopted CPA (more than half). About $2.1 billion has been raised for community preservation funding statewide.[191]

General obligation bonds are an important and popular tool for local land conservation in Minnesota as evidenced by the number of successful bond referenda. Bonds provide several advantages over pay-as-you-go funding, including the opportunity to make significant land acquisitions in the near term, presumably before the price of land increases. However, this mechanism is not always appropriate or feasible, and local governments should have the option to utilize other dedicated revenue streams, such as a dedicated property tax or local sales tax.

The state should then provide incentives to local governments to adopt these financing mechanisms by adding or increasing the local matching requirements for the state conservation trust funds and bolstering the trust funds with additional state monies. While raising taxes may seem unpopular local land conservation ballot measures have continued to enjoy strong public support across the country. In 2018, state and local voters of all political stripes approved 87 percent of the 68 parks and conservation measures on ballots nationwide, generating $7.8 billion in new parks and land conservation funding and continuing a rate of success that has been consistent since 1996. Nearly 81 percent of park and land conservation local measures placed on the ballot in Minnesota since 1996 were approved by the voters, and in many cases local property taxes were raised to pay the debt service on bonds.

Though most of the measures voted on and approved in Minnesota have taken place within the seven-county Twin Cities metro area, examples nationally show consistent support for conservation measures amongst both urban and rural areas. In 2018, Chaffee County, Colorado voters supported a sales tax increase for farmland and forest protection. The measure will generate over $1.1 million annually in a county that is over 80 percent forested federal lands. In addition, voters in all 159 counties in Georgia supported the Georgia Outdoor Stewardship Act with 83

[191] http://www.communitypreservation.org/content/cpa-overview; In 2019 the Massachusetts Legislature voted to increase the CPA fee at the state's Registries of Deeds from $20 per document to $50, bringing an additional $36 million to the CPA Trust Fund. When combined with the current registry fees and the CPA municipal property tax surcharge, annual funding for CPA will reach $225 million next year. This is a large increase dedicated funding available for parks, playgrounds, athletic fields, community gardens, both urban and rural farms, trails and acquisition conservation land and other natural climate solutions.

percent support, reallocating a portion of existing sales tax revenue to conservation. 108 of these counties are described as "rural."

Local conservation finance trends: nationally and in Minnesota

U.S. Local Conservation Finance Ballot Measures 1998-present					
Finance Mechanism	**Number of Measures**	**Pass**	**% Pass**	**Total Funds Approved**	**Conservation Funds Approved**
Property tax	1055	746	71%	$12,737,546,107	$7,705,846,189
Bond	789	663	84%	$22,058,858,488	$10,490,700,091
Sales tax	166	125	75%	$53,550,379,863	$8,701,606,734
Other	74	52	70%	$6,749,259,543	$1,233,304,282
Income tax	89	65	73%	$626,059,012	$427,159,012
Real estate transfer tax	32	27	84%	$3,138,225,154	$2,728,573,686
Total	**2205**	**1678**	**76%**	$98,860,328,167	$31,287,189,994
Minnesota Local Conservation Finance Ballot measures 1998-present					
Finance Mechanism	**Number of Measures**	**Pass**	**% Pass**	**Total Funds Approved**	**Conservation Funds Approved**
Bond	19	15	79%	$112,560,000	$87,085,000
Sales tax	2	2	100%	$5,000,000	$50,000,000
Total	**21**	**17**	**81%**	**$117,560,000**	**$92,085,000**
Source: The Trust for Public Land's LandVote database, accessed February 2019					

Nationally, voters of all political stripes continue to show strong support for local conservation funding. In 2018, local voters in 24 states approved 55 of 63 measures for parks and land conservation, generating over $3.24 billion. Property taxes continue to be the financing mechanism used most by local governments across the country

General obligation bonds

If the eighteen counties that make up the area around the Mississippi River Headwaters were each to issue bonds (assuming the average home was $200,000) with repayment costs of approximately $30 per household annually over 20 years, they would collectively generate over $132 million total, or approximately $6.6 million per year. Municipalities also are authorized to issue bonds for parks and land conservation purposes.

Mississippi Headwaters County Bond Revenue Raising Projections			
20 Year Bonds @ 5% Interest Rate			
	Annual Cost/ $200K Household		
	@ Different Bond Amounts		
COUNTY	1m	5m	10m
ANOKA	$ 0.3	$ 1.4	$ 2.7
WRIGHT	$ 0.7	$ 3.3	$ 6.5
CROW WING	$ 1.0	$ 4.9	$ 9.8
SHERBURNE	$ 1.1	$ 5.3	$ 10.7
CASS	$ 1.5	$ 7.6	$ 15.1
ITASCA	$ 1.8	$ 9.0	$ 17.9
BECKER	$ 1.9	$ 9.3	$ 18.7
MCLEOD	$ 2.8	$ 14.0	$ 28.1
BELTRAMI	$ 2.8	$ 14.1	$ 28.2
HUBBARD	$ 3.0	$ 15.1	$ 30.2
ISANTI	$ 3.2	$ 15.8	$ 31.6
MORRISON	$ 3.2	$ 16.1	$ 32.2
MEEKER	$ 3.6	$ 18.2	$ 36.4
AITKIN	$ 3.8	$ 18.8	$ 37.5
TODD	$ 4.7	$ 23.7	$ 47.3
MILLE LACS	$ 5.0	$ 25.2	$ 50.4
CLEARWATER	$ 7.9	$ 39.6	$ 79.3
WADENA	$ 9.6	$ 48.1	$ 96.2

In addition, conservation advocates could consider urging the state legislature to more broadly authorize additional local finance mechanisms, such as a local option sales tax. Such authority would provide local governments with a wide array of tools to generate matching funds.

Local Deed Recording Fee

Ten Minnesota counties participate in the Agricultural Land Preservation Program. The central feature of the program is the "agricultural preserve," a restrictive covenant on qualifying land limiting its use to agriculture or forestry. The Agricultural Land Preservation Program statute (Minn. Stat. Ch. 40A) makes it possible for owners of qualifying land to receive property tax credits of $1.50 per acre per year, and other benefits as well, in return for agreeing to preserve their farms for long-term agricultural use. Property tax credits are funded by a surcharge on mortgage and deed recordings.

These counties impose a fee of $5 per transaction on the recording or registration of a mortgage or deed that is subject to tax under Minnesota Statutes, Chapter 287.[192] Based on data from the ten counties there were around 225,000 documents upon which Minnesota's local deed fee was paid.

The total collections for FY18 was almost $1.2 million. It is estimated that collections could double if the fee was extended to all 87 counties in Minnesota.[193] That amount is split 50/50 between the state general fund and the counties to offset losses from this property tax credit program. When the fee was created in 1986 metro-counties were required to collect and other counties were given the option to be pilot counties.

Region	County	2017 Acres Enrolled	2016 County Conservation Account Revenue	Credits Reimbursed From County Account	Tax Credits Amount Left in County Conservation Account	Reimbursed From State Conservation Fund and State General Fund
Greater MN	Waseca	74,123	3,410	3,410	0	107,771
Greater MN	Winona	39,623	7,803	7,803	0	51,630
Greater MN	Wright	9,201	30,830	13,658	17,172	0
Metropolitan	Anoka	1,210	66,873	4,861	62,012	0
Metropolitan	Carver	106,895	24,122	24,122	0	359,717
Metropolitan	Dakota	71,281	80,865	80,865	0	175,087
Metropolitan	Hennepin	12,581	221,110	195,495	25,615	0
Metropolitan	Ramsey	0	76,375	0	76,375	0
Metropolitan	Scott	8,661	31,113	31,113	0	9,697
Metropolitan	Washington	7,491	54,698	26,757	27,941	0
Total		331,066 acres	$597,199	$388,084	$209,115	$703,902

Source: Minnesota Department of Revenue

Excess proceeds may be spent on agricultural land preservation and conservation planning, soil conservation activities and enforcement of soil loss ordinances; incentives for landowners who create exclusive agricultural use zones; or payments to municipalities within the county for these purposes.[194]

No other counties have shown interest in signing into the program. Most Minnesota counties follow the planning elements of the program but have expressed concern about the long-term funding of

[192] The only mortgage or deed instruments not subject to the County Conservation Fee would be those exempt under Minnesota Statues, sections 287.04 and 287.22. Information provided by the Minnesota Department of Revenue
[193] Personal communication with Minnesota Department of Revenue
[194] Minn. Stat. § 40A.152, Subd. 3.

the program. Currently, if the county and state conservation fund run short, funds are taken from the state general fund to make up the shortage.

There is an opportunity to extend this program throughout the Mississippi Headwaters to further natural climate solution efforts:

- More counties would have the incentive to enroll their land
- The fee could be increased with the extra revenue used for natural climate solutions
- The current portion of the fee that goes to the state treasury could instead be retained locally for natural climate solutions.

The deed recording fee is a mechanism used for conservation in many states and given that this program is already established, and the revenue used for conservation, this seems like an option to investigate further.

NEXT STEPS

The next steps should include narrowing funding options to those that match the needs identified Minnesota and its partners, as well as testing voter attitudes toward a specific set of funding and policy proposals. The Trust for Public Land recommends conducting a public opinion survey that tests ballot language, tax tolerance, and/or program priorities of voters in throughout the state.

APPENDIX

Minnesota Department of Natural Resources[195]

The Department of Natural Resources (DNR) uses state capital bond funds, federal grants, licenses, fees, private donations, Environment and Natural Resource Trust Fund proceeds, and general fund allocations to acquire land and conservation easements for Aquatic Management Areas, Scientific and Natural Areas, State Parks, Wildlife Management Areas, Native Prairie Bank, State Recreation Areas, State Forests, Forest Legacy conservation easements, State Trails, and Water Access.

The state of Minnesota budgets on a two-year (biennial) funding cycle. DNR's FY18-19 biennial budget is $1.1 billion. DNR's divisions include: 1) Lands and Minerals, 2) Ecological and Water Resources, 3) Forestry, 4) Parks and Trails, 5) Fish and Wildlife, 6) Enforcement, and 7) Operation Support. Parks and Trails has the largest budget ($250.4 million) and accounts for 22 percent of DNR's total spending.

Most of DNR's fiscal activity occurs in four primary funds: General Fund (26 percent), Game and Fish Fund (20 percent), Natural Resources Fund (18 percent), and Outdoor Heritage Fund (15 percent). Together, these account for 79 percent of DNR's spending.

The General Operating Fund

State General Fund dollars play an important role in DNR's budget, because they are its most flexible funding source. They allow DNR to budget for its general operating needs as well as respond quickly to emergencies and urgent, unexpected needs.

The FY18-19 General Fund budget is $293.7 million, which is less than 1 percent of the total General Fund spending for the state. Fifty-eight percent of these dollars or $171.4 million are direct appropriations for managing of the state parks and trails, minimizing the spread of invasive species, managing the state's forests and lands, protecting water, providing outdoor recreation opportunities, and enforcing natural resource laws. Direct appropriations are set by law and state agencies cannot spend more than the amount appropriated.

The remaining 42 percent are open appropriations that are available up to the amount needed to meet a statutory requirement.

The Game and Fish Fund

The Game and Fish Fund (GFF) supports management, monitoring, and protection of fish and wildlife resources as well as the enforcement of game and fish laws. The Fish and Wildlife Division accounts for 70 percent of GFF expenditures, over half of which is spent on compensation. The fund is sourced by a variety of revenues that directly relate to the programs they support. Sales of hunting and fishing licenses contributed 43 percent of GFF total revenue in FY17. GFF also receives federal funds through the federal sport fish and wildlife federal grants. DNR

[195] https://files.dnr.state.mn.us/aboutdnr/budget/fy18-19-biennial-op-budget.pdf

estimates these funds at $59.9 million this biennium. Together, these two funding sources combined contribute about 86 percent of the fund this biennium.

The Natural Resources Fund

The Natural Resources Fund (NR) supports the development and maintenance of Minnesota's natural resources and the enforcement of natural resource laws. NR expenditures make up 18 percent of DNR's total budget. This fund consists of 20 accounts that are all dedicated for a specific purpose ranging from water-based recreation to forest management to state parks. The two largest accounts are water recreation and state parks, which make up 19 percent and 18 percent of the Natural Resources Fund budget respectively.

The Parks and Trails Division accounts for 56 percent of NR spending, over half of which is spent on compensation. The two largest revenue sources to the fund include the 1) state's gasoline tax and 2) license and registration fees—together these contributed 39 percent of the fund's resources in FY17.

Constitutionally Dedicated Funding

In 1988 and 2008, Minnesota voters approved constitutional amendments that established dedicated funding for environmental, recreational, and cultural purposes. In November 1988, Minnesota voters passed a constitutional amendment that permanently established the Environment and Natural Resources Trust Fund. The Trust Fund supports projects with the public purpose of protection, conservation, preservation, and enhancement of the state's air, water land, fish, wildlife, and other natural resources. The Trust Fund receives support through lottery proceeds for land conservation. In November 1990, voters approved dedication of 40 percent of net lottery proceeds to the Trust Fund through December 2000. In 1998 voters passed another constitutional amendment extending the dedication through December 2024. In November 2008, Minnesota voters approved the Clean Water, Land and Legacy Amendment, which dedicated a sales tax increase of 3/8 percent (0.375) to clean water, natural areas, parks, arts education and history. At the time of its passage, it was the largest conservation finance ballot measure in American history. The total conservation funding over 25 years is an estimated $5.5 billion, or about $220 million annually.

DNR receives funding from four of the five constitutionally dedicated funds. These include the Environment and Natural Resources Trust Fund (ENRTF) and the following Legacy funds: Outdoor Heritage Fund, Clean Water Fund, and Parks and Trails Fund. These funding sources are unique, because 1) the projects funded are one-time in nature rather than ongoing operations and 2) Outdoor Heritage Fund and ENRTF are appropriated annually rather than biennially. In FY18-19, these funds account for 25 percent of DNR's budget.

Environment and Natural Resources Trust Fund

DNR uses these dollars to conserve and enhance Minnesota's natural resources including air, water, fish, and wildlife. The Legislative-Citizen Commission on the Minnesota Resources (LCCMR) committee governs this fund and makes annual funding recommendations to the legislature. DNR received $46.3 million from this fund in FY18-19 and 4 percent of its overall biennial budget. Trust Fund proceeds are appropriated to individual projects by the Minnesota

legislature, based upon recommendations of the Legislative-Citizen Commission on Minnesota Resources (LCCMR). LCCMR members review, evaluate, and rank all submitted proposals and the Commission as a whole then decides upon a selection of proposals to recommend for funding. Since 1991, the ENRTF has provided approximately $500 million to approximately 1,000 projects around the state.[196]

Outdoor Heritage Fund

Outdoor Heritage Fund dollars are used to restore, protect and enhance wetlands, prairies, forest and habitat for fish, game, and wildlife. The Lessard-Sams Outdoor Heritage Council (LSOHC) oversees this fund and makes recommendations to the legislature annually. DNR received $167.2 million from this fund in FY18-19, which accounts for 70 percent of its Legacy funds and 15 percent of its biennial budget.[197] Of this, $119.3 million is for pass-through grants to local governments and external organizations.

Between 2009 and June 2015, the program has acquired more than 268,000 acres and expended more than $230 million through both fee and easement purchases. A portion of dollars appropriated are made available through a small grants program administered by the Minnesota Department of Natural Resources.

Clean Water Fund

Clean Water Funding is used to protect, enhance, and restore water quality in surface and groundwater. The Clean Water Council as well as an interagency committee of state agencies makes funding recommendations to the governor. DNR received $16.9 million from this fund for FY18-19, which accounts for 7 percent of its Legacy funds and 2 percent of its overall biennial budget. Between 2011 and 2015, this fund spent nearly $15 million on land acquisition through the Board of Water and Soil Resources RIM easements.

Parks and Trails Fund

DNR uses this Legacy funding to support parks and trails of regional or statewide significance. The Greater Minnesota Regional Parks and Trails Commission provides recommendations to the legislature for grants funded by the parks and trails fund to counties and cities outside the seven-county metropolitan area. Land acquisitions made through the Parks and Trails fund will acquire within the boundaries of state parks and within state trail corridors. DNR received $54.3 million from this fund, which accounts for 23 percent of its Legacy funds and 5 percent of its biennial budget. Between 2012 and 2016, the Fund supported more than $7.3 million on land acquisition projects.

Reinvest in Minnesota

The Reinvest in Minnesota (RIM) conservation easement program permanently protects habitat and water resources. It is primarily funded through legislative bonding, the Clean Water Fund, and Outdoor Heritage Fund. It is administered by the Board of Water and Soil Resources.

[196] https://www.legacy.mn.gov/environment-natural-resources-trust-fund
[197] https://files.dnr.state.mn.us/aboutdnr/budget/fy18-19-graph.pdf

Re-Invest in Minnesota Critical Habitat Match program is administered by the MN DNR for the protection of wildlife habitat. It is funded by legislative bonding, license plate sales and private donations. It is also funded by donations from individuals matched by state funds, including license plate revenues. Vehicle owners may choose to purchase critical habitat license plates to benefit the RIM Critical Habitat Program.

Metropolitan Council

The Metropolitan Council, the planning agency serving the Twin Cities seven-county metropolitan area, works with ten regional park implementing agencies – six counties, two cities, and two park districts - to award grants to finance land acquisition and development of the parks system, among other essential regional services. The Metro Parks and Open Space Commission, with seven appointed citizen members, recommends policies and funding allocations to the full council. Some grants for land acquisition only partially finance the acquisition cost. Funds for Council grants include a combination of regional bonds issued by the Council, state bonds, user fees for wastewater treatment and transit service, a seven-county property tax, and interest earnings.

The Metropolitan Council has authority to use part of its levy for up to $40M for land protection. It typically does not bond to its full capacity because as a policy, it requires a 60 percent match from other sources including state bonding (roughly $4M/year for 10-year average), Environment and Natural Resources Trust Fund, Legacy Fund and local sources, and frequently the legislature does not provide the full match or the Met Council decides not to increase its levy to use the full amount. In addition to a Capital Improvement Program, the Met Council has created an emergency acquisition category for land acquisition projects bringing 60 percent non-regional funds, which can later be reimbursed in part with state match. This fund has generally received less than $2M in a biennium.

Previous Sources of Funding

Between 1963 and 2003, The Minnesota Future Resource Fund, which received cigarette tax revenue, funded natural resource conservation, as well as park and trail development. This 2-cent pack cigarette tax yields approximately $13-16 million per biennium. While the funding was shifted to the state General Fund in 2003, the authority for this program remains in statute.

For more information please contact:

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